

Technical Specification 43-30
Issue 2: 1981

Low Voltage Overhead Lines on
Wood Poles

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Engineering Directorate
Energy Networks Association
18 Stanhope Place
Marble Arch
London
W2 2HH

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AMENDMENT 1, 1993

Technical Specification 43-30 Issue 2, 1981

Low voltage overhead lines on wood poles

Page 7, FOREWORD Insert new paragraph at end of this section:

"This Technical Specification has been amended to introduce the changes necessary to ensure that it meets the requirements of Clause 11 of the Utilities Supply and Works Contracts Regulations 1992 (S.I. 1992 No. 3279)"

Page 7, SCOPE Add the following text:

"Since this specification is based on, The Electricity (Overhead Lines) Regulations 1970, S.I. 1970 No 1355 which have been replaced by: "The Electricity Supply Regulations 1988, S.I. 1988 No 1057" and amendments, the users of this specification should ensure that overhead lines designed to the requirements of this specification comply with the later Regulations."

Page 9. REFERENCES

Delete the complete clause and insert the following:

BS 125	Hard-drawn copper and copper-cadmium conductors for overhead power transmission purposes
BS 215	Aluminium conductors and aluminium conductors, steel-reinforced for overhead power transmission: Part 1, 'Aluminium stranded conductors' (Associated Standard IEC 1089)
BS 1990	Wood poles for overhead power and telecommunication lines
BS 6346	PVC-insulated cables for electricity supply
BS 6485	PVC-covered conductors for overhead power lines

- EA TS 09-7 PVC-insulated concentric cables with stranded copper or solid aluminium phase conductors and copper concentric neutral conductors
- EA TS 43-88 Treatment of wood poles and associated timber for overhead lines
- EA TS 43-90 Anti-climbing devices and safety signs for high voltage lines up to and including 400 kV
- EA TS 43-91 Stay strands and stay fittings for overhead lines
- EA TS 43-92 Conductor terminations, joints and insulator binds for overhead lines up to and including 132 kV
- EA TS 43-93 Line insulators
- EA TS 43-95 Steelwork for overhead lines
- EA TS 43-96 Fasteners and washers for wood pole overhead lines
- S.I. 1988 The Electricity Supply Regulations 1988
No 1057
- PO 1 (1975): Post Office Memorandum A80e Protection of Post Office lines from contact with low or power lines medium voltage

General

All references in the text to "ESI Standard" to be replaced by "EA Technical Specification"

Page 12 Clause 5.2 Cables

Delete the reference to: "BS 6004" (PVC-insulated cables (non-armoured) for Electric power and lighting).

Replace with: "BS 6346" (PVC-insulated cables for electricity supply.)

Page 12 Clause 5.3 Wood Poles

Delete sub clause (i) and replace with the following text: "Poles shall comply with the requirements of Technical Specification EA TS 43-88."

Delete sub clause (v)

Page 14 Clause 6.8 (i)

To be replaced by "... with The Electricity Supply Regulations 1988, schedule 2 columns 2 and 3."

Page 14 Clause 6.9

Replace "the Post Office" by "BT"

Page 58 Fig 22

Delete reference: "EA TS 43-97" (Notices and marker posts for H.V. lines upto and including 132 kV)

Replace with: "EA TS 43-90" (Anti-climbing devices and safety signs for H.V. lines up to and including 400 kV)

Delete reference to "LV and HV earthing to ESI 43-89"

Page 59 Fig 22

Remove page 59 (A duplicate of page 58.)

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LOW VOLTAGE OVERHEAD LINES ON WOOD POLES

FOREWORD

Since 1962, the Electricity Supply Industry's low voltage overhead lines have been erected to British Electricity Board Specification L1. With the introduction of metrication and the revision of the Electricity (Overhead Lines) Regulations 1970, a need has been created for a revised metric specification. This Standard has been prepared to replace BEB Specification L1 but, because of the proven reliability of the latter, certain aspects have been retained. The opportunity has also been taken to rationalize where necessary.

Although the kilonewton is recognized as the approved metric unit of force, it is intended to continue using kilogram-force to take account of the practical difficulties associated with the calibration of lifting and pulling equipment.

It has been agreed with British Telecom that the permanent protection of stays above the insulator should cease. Fig. 18 is therefore no longer applicable and has been removed from the Standard.

1. SCOPE

This Standard covers single- and three-phase low voltage overhead lines on wood poles (including street lighting switch wires, service spans and lead-ins, pole mounted transformer connections and earthing arrangements) employing the following conductors and cables.

Distribution Lines

Hard-drawn copper conductor	32 mm ² and 70 mm ²
Hard-drawn aluminium conductor	50 mm ² and 100 mm ²

Service Lines

Hard-drawn copper conductor	16 mm ²
Hard-drawn aluminium conductor	25 mm ²

Service Spans

Hard-drawn copper conductor	16 mm ²
Hard-drawn aluminium conductor	25 mm ²
PVC concentric cable with stranded copper phase conductor and copper concentric neutral conductor	16 mm ² and 25 mm ²

Service Lead-in

Plain annealed copper insulated and sheathed cable	16 mm ² , 25 mm ² , 35 mm ² and 70 mm ²
Soft-drawn aluminium insulated and sheathed cable	25 mm ² , 35 mm ² , 50 mm ² and 95 mm ²

Transformer Connections

Plain annealed copper insulated and sheathed cable	35 mm ² , 70 mm ² , 95 mm ² and 120 mm ²
Soft-drawn aluminium insulated and sheathed cable	50 mm ² , 95 mm ² , 120 mm ² and 185 mm ²

Earthing Arrangements

Hard-drawn copper conductor, PVC-covered	16 mm ² and 32 mm ²
Plain annealed copper conductor, insulated and sheathed	16 mm ² , 25 mm ² and 35 mm ²
Soft-drawn aluminium conductor, insulated and sheathed	25 mm ² and 50 mm ²

Guard Wires

Hard-drawn copper conductor	16 mm ²
Hard-drawn aluminium conductor	25 mm ²

2. DEFINITIONS

For the purposes of this Standard the following definitions apply:

Intermediate Support — A support in a line where the conductors are continuous and supported on reel insulators, without insulator brackets, with up to 15° line deviation.

Intermediate Angle Support — A support in a line where the conductors are continuous and supported by reel insulators in insulator brackets with line deviations above 15° up to 60°.

Section Angle Support — A support in a line on which the conductors are made off on both sides of the pole on to reel insulators in insulator brackets.

Terminal Support — A support on which the conductors are terminated on reel insulators in insulator brackets on one side of the pole only.

Span — The horizontal distance between adjacent supports.

Wind Loading Span — The wind loading span associated with any support is half the sum of the spans adjacent to the support.

Average Span — The arithmetical average of a number of spans in a line or section of line.

Basic Span — The span length adopted for sag/tension calculations.

Recommended Span — The average span length in any section to which the line should be planned.

Maximum Span — The maximum permitted length of any span.

Sag — The projected vertical distance under any condition of conductor loading, between the conductor and a straight line joining adjacent supporting points, measured mid-span.

Distribution Line — A line fed from a transformer, providing a supply to more than one consumer, separately tapped.

Service Line — A line feeding one consumer from a distribution line or transformer, consisting of one span or more excluding the last span to the consumer's premises (see Service Span).

Service Span — The last span to a consumer terminating on the consumer's building.

Offset Arrangement — An arrangement where the conductors are supported offset from the pole by means of a bracket.

Flying Stay — A means of providing a horizontal load to an angle support where it is impracticable to use the conventional method of staying.

Guard Wire — An earthed conductor that is suspended above the power conductors to provide protection to overrunning Post Office wires in the event of their failure.

3. REFERENCES

This Standard makes reference to the following documents:

- BS 125, 'Hard-drawn Copper and Copper-Cadmium Conductors for Overhead Power Transmission Purposes'.
- BS 215, 'Aluminium Conductors and Aluminium Conductors, Steel-reinforced for Overhead Power Transmission'.
Part 1, 'Aluminium Stranded Conductors'.

BS 913, 'Wood Preservation by means of Pressure Creosoting'.
BS 1990, 'Wood Poles for Overhead Lines (Power and Telecommunication Lines)'.
BS 3452, 'Copper/Chrome Water-borne Wood Preservatives and their Application'.
BS 4072, 'Wood Preservation by means of Water-borne Copper/Chrome/Arsenic Compositions'.
BS 6004, 'PVC-insulated Cables (Non-armoured) for Electric Power and Lighting'.
BS 6485, 'PVC-covered Conductors for Overhead Power Lines'.
ESI Standard 09—7, 'PVC Insulated Concentric Cables with Stranded Copper or Solid Aluminium Phase Conductors and Copper Concentric Neutral Conductors'.
*ESI Standard 43—89, 'Auxiliary Equipment Erected on Overhead Line Supports including Earthing and Bonding'.
ESI Standard 43—90, 'Anti-climbing Devices for H.V. Lines up to and including 132 kV'.
ESI Standard 43—91, 'Stay Strands and Stay Fittings for Overhead Lines'.
ESI Standard 43—92, 'Conductor Terminations, Joints and Insulator Binds for Overhead Lines up to and including 132 kV'.
ESI Standard 43—93, 'Line Insulators'.
ESI Standard 43—95, 'Steelwork for Overhead Lines'.
*ESI Standard 43—96, 'Fasteners and Washers for Wood Pole Overhead Lines'.
*ESI Standard 43—97, 'Notices and Marker Posts for H.V. Lines up to and including 132 kV'.
The Electricity (Overhead Lines) Regulation 1970, SI 1355.
PO Memorandum A80, 'The Protection of Post Office Lines from Contact with Low or Medium Voltage Power Lines'.
PO Memorandum A2772, 'Post Office and Electricity Board's Memorandum of Agreement'.

*In course of preparation.

4. DESIGN DATA

This Standard is designed in accordance with the requirements of the Electricity (Overhead Lines) Regulations 1970.

4.1 Conductor Tensions

Actual maximum working tensions and factors of safety are shown below:

Conductor Size and Type	Situation	Max. Working Tension		Factor of Safety
		kgf	kN	
16 mm ² concentric cable	Service span	132	1.3	
25 mm ² concentric cable	Service span	132	1.3	
16 mm ² copper bare and PVC	Service span	132	1.3	5.07
25 mm ² aluminium bare and PVC	Service span	132	1.3	3.33
16 mm ² copper bare and PVC	Service line	220	2.16	3.05
25 mm ² aluminium bare and PVC	Service line	220	2.16	2.00
32 mm ² copper bare and PVC	Distribution line	454	4.45	2.86
50 mm ² aluminium bare and PVC	Distribution line	422	4.15	2.00
70 mm ² copper bare and PVC	Distribution line	612	6.00	4.48
100 mm ² aluminium bare and PVC	Distribution line	612	6.00	2.67

Notes: 1. Conductor design tensions take into account the following conditions:

- (i) The minimum factors of safety on conductors shall be 2.0 on their calculated breaking loads.
- (ii) The maximum working tension shall not exceed 454 kgf (4.45 kN) for 32 mm² copper or 612 kgf (6.00 kN) for 70 mm² copper and 100 mm² aluminium to enable extensions to be carried out to existing BEB Specification L1 lines without introducing imbalance.
- (iii) The maximum working tension of service spans shall not exceed 132 kgf (1.3 kN). This figure has been chosen as the maximum acceptable load that can be applied to a single point fixing to a building.

- 2. In the preparation of sag/tension tables, basic span lengths of 50 m for distribution and service lines, and 20 m for service spans have been adopted. (Tables 7—28.)

4.2 Supports

Intermediate and section support sizes (see Tables 2—4) are related to loadings imposed by PVC-covered conductors, the loading point being considered to be 600 mm below the pole top. Stayed supports are designed in accordance with crippling loads imposed by PVC-covered conductor loadings (see Charts 8—10). The strength of supports are derived from information contained in BS 1990 with respect to red fir poles. The minimum design factor of safety for supports is 2.5 under the worst loading conditions stipulated by the Electricity (Overhead Lines) Regulations 1970.

4.3 Steelwork

Steelwork specified to ESI Standard 43—95 is designed to withstand the maximum working tensions tabled in 4.1 above.

5. MATERIALS

All components used in the construction of lines to this Standard shall comply with the British Standards and/or ESI Standards indicated below:

5.1 Conductors

5.1.1 *Types Specified*

- | | | |
|---------------------------------------|---|----------|
| (i) Hard-drawn copper conductors | — | BS 125. |
| (ii) Hard-drawn aluminium conductors | — | BS 215. |
| (iii) PVC-covered conductors (Type 8) | — | BS 6485. |

5.1.2 *Greasing of Wires*

During stranding, all inner wires shall be coated with an approved grease in such manner that the interstices between wires are effectively filled with grease within the circle circumscribing the penultimate layer of the complete conductor.

For extra protection of bare aluminium conductors in corrosive areas and when specified by the purchaser, all wires shall be greased and the conductor subsequently wiped to remove excess grease from the outer surface whilst maintaining grease filling in the inter-wire crevices.

5.2 Cables

- (i) PVC-insulated and sheathed cables shall comply with BS 6004.
- (ii) PVC-insulated concentric cables with stranded copper phase conductors and copper concentric neutral conductors shall comply with ESI Standard 09—7.

5.3 Wood Poles

- (i) Poles shall be best grade European redwood with a minimum ultimate extreme fibre stress of 53.8 N/mm², complying with the requirements of BS 1990.
- (ii) The poles shall be fabricated to Fig. 1, all holes being drilled to template, the centre line of the template being placed along the effective centre line of the pole.
- (iii) Pole markings shall be made on the same side of the pole as that on which the drilling templates have been used.
- (iv) Pole tops shall be evenly quartered, the angle of apex being 120°.
- (v) When fully seasoned and fabricated, poles shall be creosoted in accordance with BS 913 by the Rueping Process. An alternative method of wood preservation may be applied in accordance with BS 3452 or BS 4072 if specified by the purchaser.

5.4 Insulators

Reel insulators and coachscrew insulators shall comply with ESI Standard 43—93.

5.5 Conductor Terminations, Joints and Insulator Binds

Conductor terminations, joints and binders shall comply with ESI Standard 43—92.

5.6 Stay Strand

Stay strand of the following types shall comply with ESI Standard 43—91.

- (i) Grade 700 — 4/4.0 mm and 7/4.0 mm.
- (ii) Grade 1150 — 7/2.65 mm and 7/3.25 mm.

5.7 Stay Components

The following stay components shall comply with ESI Standard 43—91:

- (i) Stay make-offs.
- (ii) Stay rods.
- (iii) Stay insulators.
- (iv) Stay blocks.
- (v) Stay thimbles.

5.8 Fasteners and Washers

Fasteners and washers shall comply with ESI Standard 43—96.

6. TECHNICAL REQUIREMENTS FOR THE ERECTION OF LINES

6.1 General

Supports shall be erected in accordance with Figs. 2—8.

6.2 Maximum and Recommended Span Lengths

Maximum and recommended span lengths are specified in Table 1.

6.3 Wind Loading Spans

Wind loading spans (based on loadings imposed by PVC-covered conductors) to be used for bare and PVC-covered conductors are given in Tables 2—4.

6.4 Staying Arrangements and Strut Loading for Angle and Terminal Supports

- (i) Staying arrangements are given in Table 6 which should be read in conjunction with the associated notes.
- (ii) Stay assemblies shall be to ESI Standard 43—91.
- (iii) Where a change of conductor type or size occurs at a section angle support, stays should be fitted to balance the resultant loading imposed on that support.
- (iv) The strut loading imposed upon section angle and terminal supports shall be obtained from Charts 8—10 as appropriate. These loadings shall then be related to the safe working loads of the poles by reference to Table 5.

6.5 Conductor Erection Sags and Tensions

Conductor sags and tensions are given in Tables 7—28.

6.6 Use of PVC-covered Conductors

Technical data are provided in this Standard for the erection of full section(s) and single spans of PVC-covered phase conductors. Bare neutral conductors should in all cases be regulated to the PVC-covered conductors.

6.7 Conductor Configuration and Spacing

6.7.1 Service Lines and Distribution Lines

- (i) The nominal arrangement of conductors for a three-phase, four-wire service line or distribution line where a switch wire is not required shall be as follows:

Red No. 1 (TOP).
Yellow No. 2.
Blue No. 3.
Neutral No. 4.

Where a switch wire is required, this shall occupy position No. 4 and the neutral be placed in the fifth position below the switch wire.

- (ii) Conductor spacing shall be 300 mm.

6.7.2 Service Spans

- (i) The PVC-covered phase conductors shall always be positioned below the neutral conductor on open wire services.
- (ii) Conductor spacing shall be 150 mm.

6.8 Minimum Height of Line Conductors and Clearances

- (i) The minimum height of line conductors shall comply with the Electricity (Overhead Lines) Regulations 1970, Schedule 1, columns 2 and 3.
- (ii) The data given in this Standard are for conductors operating at a likely maximum temperature of 50°C.
- (iii) An additional clearance of 250 mm shall be allowed for concentric service cables to compensate for conductor creep.

6.9 Relationship with Other Bodies

Where lines to this Standard are erected over or alongside the plant of the Post Office or British Rail, the provisions agreed shall be complied with. Additionally, the requirements of the British Waterways Board should be noted.

Table 1 — Recommended and Maximum Span Lengths

Conductor or Cable Size		Recommended Span (m)	Maximum Span (m)
16 mm ² 25 mm ²	Copper Concentric Cable (Service Spans)	20	30
16 mm ² 25 mm ²	Copper (Service) Aluminium (Spans)	20	30
16 mm ² 25 mm ²	Copper (Service) Aluminium (Lines)	50	50
32 mm ² 50 mm ² 70 mm ² 100 mm ²	Copper Aluminium Copper Aluminium } Distribution Lines	50	60

**Table 2 — Wind Loading Spans for Intermediate and Section Poles
16 mm² Copper and 25 mm² Aluminium Bare and PVC-covered Conductors**

Pole Details			Wind Loading Spans (m)			
Length (m)	Planting Depth (m)	Dia. at 1.5 m from butt (mm)	2 Conductors	3 Conductors	4 Conductors	5 Conductors
8.5	1.5	180 (L) 185 190 195	60	60	58 60	49 55 59 60
9.0	1.5	180 (L) 185 190 195 200	60	60	56 60	45 49 55 59 60
9.5	1.5	185 (L) 190 195 200 205	60	60	56 60	45 49 55 59 60
10.0	1.5	185 (L) 190 195 200 205	60	60	52 56 60	42 46 50 54 60
10.5	1.5	190 (L) 195 200 205 210	60	60	52 58 60	42 46 50 56 60
11.0	1.8	195 (L) 200 205 210 215	60	60	53 59 60	42 47 51 56 60

(Cont'd)

Table 2 (Cont'd)

Pole Details			Wind Loading Spans (m)			
Length (m)	Planting Depth (m)	Dia. at 1.5 m from butt (mm)	2 Conductors	3 Conductors	4 Conductors	5 Conductors
11.5	1.8	200 (L)	60	60	54 60	43 48 53 56 60
		205				
		210				
		215				
		220				
12.0	1.8	200 (L)	60	60	50 56 60	40 45 48 53 57 60
		205				
		210				
		215				
		220				
13.0	1.8	225	60	60	51 58 60	41 46 50 54 58 60
		210 (L)				
		215				
		220				
		225				
		230				
		235				

Notes: 1. (L) against a pole diameter indicates that the pole is the standard light pole in accordance with BS 1990.
2. Poles are assumed to have the minimum top diameter in accordance with BS 1990.

Table 3 — Wind Loading Spans for Intermediate and Section Poles
32 mm² Copper and 50 mm² Aluminium Bare and PVC-covered Conductors

Pole Details			Wind Loading Spans (m)			
Length (m)	Planting Depth (m)	Dia. at 1.5 m from butt (mm)	2 Conductors	3 Conductors	4 Conductors	5 Conductors
8.5	1.5	180 (L)	60	60	55 60	44
		185				48
		190				52
		195				57
		200				60
9.0	1.5	180 (L)	60	60	50 54 60	40
		185				44
		190				48
		195				52
		200				57
9.5	1.5	205				60
		210				60
		215				60
		220				60
		225				60
10.0	1.5	185 (L)	60	59 60	46 50 55 60	37
		190				40
		195				44
		200				48
		205				52
10.5	1.5	210				57
		215				60
		220				60
		225				60
		230				60
11.0	1.8	190 (L)	60	60	46 51 56 60	37
		195				41
		200				45
		205				49
		210				53
11.5	1.8	215				57
		220				60
		225				60
		230				60
		235				60

(Cont'd)

Table 3 (Cont'd)

Pole Details			Wind Loading Spans (m)			
Length (m)	Planting Depth (m)	Dia. at 1.5 m from butt (mm)	2 Conductors	3 Conductors	4 Conductors	5 Conductors
12.0	1.8	200 (L)	60	59	45	36
		205		60	49	39
		210			54	43
		215			58	46
		220			60	50
		225				53
		230				57
		235				60
13.0	1.8	210 (L)	60	60	47	38
		215			51	41
		220			55	44
		225			59	48
		230			60	51
		235				54
		240				58
		245				60

Notes: 1. (L) against a pole diameter indicates that the pole is the standard light pole in accordance with BS 1990.
 2. Poles are assumed to have the minimum top diameter in accordance with BS 1990.

**Table 4 — Wind Loading Spans for Intermediate and Section Poles
70 mm² Copper and 100 mm² Aluminium Bare and PVC-covered Conductors**

Pole Details			Wind Loading Spans (m)			
Length (m)	Planting Depth (m)	Dia. at 1.5 m from butt (mm)	2 Conductors	3 Conductors	4 Conductors	5 Conductors
8.5	1.5	180 (L)	60	58	45	36
		185				
		190				
		195				
		200				
		205				
		210				
9.0	1.5	180 (L)	60	53	41	33
		185				
		190				
		195				
		200				
		205				
9.5	1.5	185 (L)	60	53	41	33
		190				
		195				
		200				
		205				
		210				
10.0	1.5	185 (L)	60	49	38	30
		190				
		195				
		200				
		205				
		210				
10.5	1.5	190 (L)	60	49	38	31
		195				
		200				
		205				
		210				
		215				
11.0	1.8	195 (L)	60	51	39	31
		200				
		205				
		210				
		215				
		220				
11.0	1.8	225	60	56	43	34
		230				

(Cont'd)

Table 4 (Cont'd)

Pole Details			Wind Loading Spans (m)			
Length (m)	Planting Depth (m)	Dia. at 1.5 m from butt (mm)	2 Conductors	3 Conductors	4 Conductors	5 Conductors
11.5	1.8	200 (L)	60	52	40	32
		205		57	44	35
		210		60	47	38
		215			51	41
		220			55	44
		225			59	47
		230			60	50
		235				54
12.0	1.8	200 (L)	60	49	37	30
		205		53	41	33
		210		57	44	35
		215		60	48	38
		220			51	41
		225			55	44
		230			59	47
		235			60	51
13.0	1.8	240				54
		210 (L)	60	51	39	31
		215		55	42	34
		220		59	46	37
		225		60	49	39
		230			53	42
		235			56	45
		240			60	48
		245				51
		250				55
		255				58

Notes: 1. (L) against a pole diameter indicates that the pole is the standard light pole in accordance with BS 1990.
2. Poles are assumed to have the minimum top diameter in accordance with BS 1990.

Table 5 — Permissible Strut Loading of Poles

Pole Planting Depths (m)	Pole Lengths (m)	Pole Top Dia. (m)	Max. Strut Loading (kgf)	Pole Top Dia. (mm)	Max. Strut Loading (kgf)	Pole Top Dia. (mm)	Max. Strut Loading (kgf)	Dia. at 1.5 m (mm)	Pole Class (BS 1990)
1.5	8.5	125	1543	140	1978	150	2316	180	L
	9.0	125	1338	140	1718	150	2013	180	
	9.5	125	1232	140	1580	160	2151	185	
	10.0	125	1088	140	1396	160	1903	185	
	10.5	125	1016	140	1301	160	1770	190	
1.8	11.0	125	998	140	1282	160	1750	195	
	11.5	125	939	145	1303	165	1764	200	
	12.0	125	848	145	1177	165	1594	200	
	13.0	130	835	150	1144	170	1533	210	
1.5	8.5	150	3171	165	3907	180	4765	215	M
	9.0	150	2870	165	3533	180	4305	220	
	9.5	150	2620	165	3222	180	3922	225	
	10.0	150	2410	165	2960	185	3833	230	
	10.5	150	2231	170	2923	190	3765	235	
1.8	11.0	150	2170	170	2854	190	3689	240	
	11.5	150	2025	170	2658	190	3429	245	
	12.0	150	1899	170	2487	190	3204	250	
	13.0	160	1936	180	2505	200	3192	260	
1.5	8.5	190	7763	215	10229	240	13240	265	S
	9.0	190	7209	215	9471	240	12227	275	
	9.5	190	6526	215	8565	245	11605	280	
	10.0	190	5953	215	7805	245	10664	285	
	10.5	190	5467	220	7542	240	10158	290	
1.8	11.0	190	5282	220	7325	250	9910	295	
	11.5	190	4892	220	6772	250	9148	300	
	12.0	190	4553	220	6292	250	8487	305	
	13.0	195	4351	225	5953	255	7961	320	
Poles with top dia. equal to min. specified in BS 1990.			Poles with mean value of top dia.			Poles with top dia. equal to max. specified in BS 1990.			

Notes: 1. The above values of strut loading are safe working loads.
2. This table should be read in conjunction with Charts 8, 9 and 10.

Table 6 — Stay Details

Stay Strand	Stay Rod	Stay Insulator	Stay Block	MFL Assembly (kgf)	Component Determining MFL	Use
Grade 700 4/4.0 mm or Grade 1150 7/2.65 mm	439104 Screw-in	439107 Porcelain Type 1	—	1224	Stay Rod	Services and Service Lines
Grade 700 4/4.0 mm or Grade 1150 7/2.65 mm	439101 Type 1	439107 Porcelain Type 1	439103 Type 1 or 439102	3589	Strand	Service Lines and Distribution Lines
				4528	Strand	
Grade 700 7/4.0 mm or Grade 1150 7/3.25 mm	439101 Type 1	439107 Porcelain Type 1	439103 Type 1 or 439102	6281	Strand	Distribution Lines
				6628	Insulator	

- Notes:** 1. **Terminal stays.** Terminal stays shall be in accordance with the corresponding requirements for 60° angles of deviation, as shown in Charts 2—7.
2. **Strut Loading of Poles.** Poles suitable for combinations of stay slopes and angles of deviation are given in Table 5.
3. **Stay Spread and Staying Requirements.** This information is given in Charts 1—7.
4. **Stay Assembly Strengths.** For the basis of staying requirement for distribution lines, as shown in Charts 2—7, stay assemblies have been divided into two groups:

- | | | |
|-------------------|---|---------------------------------------|
| Group 1 | — | 4/40, 700 N/mm ² grade. |
| (Rating 3589 kgf) | — | 7/2.65, 1150 N/mm ² grade. |
| Group 2 | — | 7/4.40, 700 N/mm ² grade. |
| (Rating 6281 kgf) | — | 7/3.25, 1150 N/mm ² grade. |

**Table 7 — Erection Sag/Tension Tables for 16 mm²
Concentric Cable O/H Service Spans**

Temp °C	Tension (kgf)	Sag (m) for Span Length (m)		
		10	20	30
0	56.4	0.10	0.40	0.90
10	54.0	0.10	0.42	0.94
20	51.8	0.11	0.43	0.98
30	49.9	0.11	0.45	1.02
40	48.1	0.12	0.47	1.05
50	46.6	0.12	0.48	1.09

**Table 8 — Erection Sag/Tension Tables for 25 mm²
Concentric Cable O/H Service Spans**

Temp °C	Tension (kgf)	Sag (m) for Span Length (m)		
		10	20	30
0	102.6	0.08	0.32	0.71
10	95.4	0.09	0.34	0.77
20	89.4	0.09	0.36	0.82
30	84.0	0.10	0.39	0.87
40	79.2	0.10	0.41	0.92
50	75.6	0.11	0.43	0.97

**Table 9 — Erection Sag/Tension Tables for 16 mm²
Copper Conductor — Bare and PVC-covered
Open Wire Service — Spans**

Temp °C	Tension (kgf)	Sag (m) for Span Length (m)				
		10.0	15.0	20.0	25.0	30.0
-5	54	0.04	0.09	0.17	0.26	0.38
0	47	0.05	0.11	0.19	0.30	0.43
5	42	0.05	0.12	0.21	0.33	0.48
10	38	0.06	0.13	0.23	0.37	0.53
15	35	0.06	0.14	0.26	0.40	0.58
20	33	0.07	0.16	0.28	0.43	0.62
25	30	0.07	0.17	0.30	0.46	0.67
30	29	0.08	0.18	0.31	0.49	0.71
35	27	0.08	0.19	0.33	0.52	0.75
40	26	0.09	0.20	0.35	0.55	0.79
45	25	0.09	0.21	0.37	0.57	0.82
50	24	0.10	0.22	0.38	0.60	0.86

**Table 10 — Erection Sag/Tension Tables for 25 mm²
Aluminium Conductor — Bare and PVC-covered
Open Wire Service — Spans**

Temp °C	Tension (kgf)	Sag (m) for Span Length (m)				
		10.0	15.0	20.0	25.0	30.0
-5	56	0.02	0.05	0.10	0.15	0.22
0	44	0.03	0.07	0.12	0.19	0.27
5	35	0.04	0.09	0.15	0.24	0.34
10	29	0.05	0.10	0.19	0.29	0.42
15	25	0.05	0.12	0.22	0.34	0.49
20	22	0.06	0.14	0.25	0.39	0.56
25	19	0.07	0.16	0.28	0.44	0.63
30	18	0.08	0.17	0.31	0.48	0.69
35	16	0.08	0.18	0.33	0.51	0.74
40	15	0.09	0.20	0.35	0.55	0.80
45	15	0.09	0.21	0.37	0.58	0.84
50	14	0.10	0.22	0.40	0.62	0.89

Erection Sag/Tension Tables for 16 mm² Copper Conductors — Basic Span 50 mm
(Tables 11—13)

Table 11 — Bare Conductors

Temp °C	Tension (kgf)	Sag (m) for Span Length (m)								
		20	25	30	35	40	45	50	55	60
-5.0	87	0.09	0.14	0.20	0.27	0.35	0.44	0.54	0.66	0.78
0.0	81	0.09	0.15	0.21	0.29	0.38	0.48	0.59	0.71	0.85
5.0	75	0.10	0.16	0.23	0.31	0.40	0.51	0.63	0.76	0.91
10.0	70	0.11	0.17	0.24	0.33	0.43	0.55	0.68	0.82	0.97
15.0	66	0.12	0.18	0.26	0.35	0.46	0.58	0.72	0.87	1.04
20.0	62	0.12	0.19	0.27	0.37	0.49	0.62	0.76	0.92	1.10
25.0	59	0.13	0.20	0.29	0.39	0.51	0.65	0.80	0.97	1.16
30.0	56	0.14	0.21	0.30	0.41	0.54	0.68	0.84	1.02	1.22
35.0	54	0.14	0.22	0.32	0.43	0.57	0.72	0.88	1.07	1.27
40.0	51	0.15	0.23	0.33	0.45	0.59	0.75	0.92	1.12	1.33
45.0	49	0.15	0.24	0.35	0.47	0.61	0.78	0.96	1.16	1.38
50.0	48	0.16	0.25	0.36	0.49	0.64	0.81	1.00	1.21	1.44

Table 12 — PVC-covered Conductors (Sections)

Temp °C	Tension (kgf)	Sag (m) for Span Length (m)								
		20	25	30	35	40	45	50	55	60
-5.0	78	0.12	0.18	0.26	0.35	0.46	0.58	0.72	0.87	1.04
0.0	74	0.12	0.19	0.27	0.37	0.49	0.62	0.76	0.92	1.10
5.0	70	0.13	0.20	0.29	0.39	0.51	0.65	0.80	0.97	1.16
10.0	67	0.13	0.21	0.30	0.41	0.54	0.68	0.84	1.02	1.21
15.0	64	0.14	0.22	0.32	0.43	0.56	0.71	0.88	1.07	1.27
20.0	61	0.15	0.23	0.33	0.45	0.59	0.74	0.92	1.11	1.32
25.0	59	0.15	0.24	0.34	0.47	0.61	0.77	0.96	1.16	1.38
30.0	57	0.16	0.25	0.36	0.49	0.63	0.80	0.99	1.20	1.43
35.0	55	0.16	0.26	0.37	0.50	0.66	0.83	1.03	1.24	1.48
40.0	53	0.17	0.27	0.38	0.52	0.68	0.86	1.06	1.28	1.53
45.0	51	0.18	0.27	0.39	0.54	0.70	0.89	1.09	1.32	1.58
50.0	50	0.18	0.28	0.41	0.55	0.72	0.91	1.13	1.36	1.62

Table 13 — PVC-covered Conductors (Single Spans)

Temp °C	Tension (kgf)	Sag (m) for Span Length (m)								
		20	25	30	35	40	45	50	55	60
-5.0	87	0.10	0.16	0.23	0.32	0.41	0.52	0.64	0.78	0.93
0.0	81	0.11	0.17	0.25	0.34	0.45	0.56	0.70	0.84	1.00
5.0	75	0.12	0.19	0.27	0.37	0.48	0.61	0.75	0.91	1.08
10.0	70	0.13	0.20	0.29	0.39	0.51	0.65	0.80	0.97	1.15
15.0	66	0.14	0.21	0.31	0.42	0.55	0.69	0.85	1.03	1.23
20.0	62	0.14	0.23	0.32	0.44	0.58	0.73	0.90	1.09	1.30
25.0	59	0.15	0.24	0.34	0.47	0.61	0.77	0.95	1.15	1.37
30.0	56	0.16	0.25	0.36	0.49	0.64	0.81	1.00	1.21	1.44
35.0	54	0.17	0.26	0.38	0.51	0.67	0.85	1.05	1.27	1.51
40.0	51	0.17	0.27	0.29	0.54	0.70	0.88	1.09	1.32	1.57
45.0	49	0.18	0.28	0.41	0.56	0.73	0.92	1.14	1.38	1.64
50.0	48	0.19	0.30	0.42	0.58	0.76	0.96	1.18	1.43	1.70

Erection Sag/Tension Tables for 25 mm² Aluminium Conductors Basic Span 50 m (Tables 14—16)

Table 14 — Bare Conductors

Temp °C	Tension (kgf)	Sag (m) for Span Length (m)								
		20	25	30	35	40	45	50	55	60
-5.0	87	0.04	0.06	0.09	0.12	0.16	0.20	0.25	0.30	0.36
0.0	73	0.05	0.07	0.11	0.14	0.19	0.24	0.29	0.36	0.42
5.0	62	0.06	0.09	0.13	0.17	0.22	0.28	0.35	0.42	0.50
10.0	52	0.07	0.10	0.15	0.20	0.27	0.34	0.41	0.50	0.60
15.0	45	0.08	0.12	0.17	0.24	0.31	0.39	0.48	0.58	0.70
20.0	39	0.09	0.14	0.20	0.27	0.35	0.45	0.55	0.67	0.80
25.0	35	0.10	0.16	0.22	0.31	0.40	0.50	0.62	0.75	0.90
30.0	31	0.11	0.17	0.25	0.34	0.44	0.56	0.69	0.83	0.99
35.0	29	0.12	0.19	0.27	0.37	0.48	0.61	0.75	0.91	1.08
40.0	26	0.13	0.20	0.29	0.40	0.52	0.66	0.81	0.99	1.17
45.0	25	0.14	0.22	0.31	0.43	0.56	0.71	0.87	1.05	1.26
50.0	23	0.15	0.23	0.33	0.45	0.59	0.75	0.92	1.12	1.33

Table 15 — PVC-covered Conductors (Sections)

Temp °C	Tension (kgf)	Sag (m) for Span Length (m)								
		20	25	30	35	40	45	50	55	60
-5.0	70	0.08	0.12	0.17	0.24	0.31	0.39	0.48	0.58	0.70
0.0	62	0.09	0.14	0.20	0.27	0.35	0.44	0.54	0.66	0.78
5.0	56	0.10	0.15	0.22	0.30	0.39	0.49	0.60	0.73	0.87
10.0	51	0.11	0.17	0.24	0.33	0.42	0.54	0.66	0.80	0.96
15.0	47	0.12	0.18	0.26	0.35	0.46	0.59	0.72	0.87	1.04
20.0	43	0.12	0.20	0.28	0.38	0.50	0.63	0.78	0.94	1.12
25.0	40	0.13	0.21	0.30	0.41	0.53	0.68	0.84	1.01	1.20
30.0	38	0.14	0.22	0.32	0.44	0.57	0.72	0.89	1.08	1.28
35.0	36	0.15	0.24	0.34	0.46	0.60	0.76	0.94	1.14	1.35
40.0	34	0.16	0.25	0.36	0.49	0.63	0.80	0.99	1.20	1.43
45.0	33	0.17	0.26	0.37	0.51	0.66	0.84	1.04	1.26	1.49
50.0	31	0.17	0.27	0.39	0.53	0.70	0.88	1.09	1.32	1.57

Table 16 — PVC-covered Conductors (Single Spans)

Temp °C	Tension (kgf)	Sag (m) for Span Length (m)								
		20	25	30	35	40	45	50	55	60
-5.0	87	0.06	0.10	0.14	0.19	0.25	0.32	0.39	0.47	0.56
0.0	73	0.07	0.12	0.17	0.23	0.30	0.37	0.46	0.56	0.66
5.0	62	0.09	0.14	0.20	0.27	0.35	0.44	0.55	0.66	0.79
10.0	52	0.10	0.16	0.23	0.32	0.41	0.53	0.65	0.78	0.93
15.0	45	0.12	0.19	0.27	0.37	0.48	0.61	0.76	0.92	1.09
20.0	39	0.14	0.22	0.31	0.42	0.56	0.70	0.87	1.05	1.25
25.0	35	0.16	0.24	0.35	0.48	0.62	0.79	0.98	1.18	1.40
30.0	31	0.17	0.27	0.39	0.53	0.69	0.87	1.08	1.30	1.55
35.0	29	0.19	0.29	0.42	0.58	0.75	0.95	1.18	1.42	1.69
40.0	26	0.20	0.32	0.46	0.62	0.82	1.03	1.28	1.54	1.84
45.0	25	0.22	0.34	0.49	0.67	0.87	1.11	1.36	1.65	1.96
50.0	23	0.23	0.36	0.52	0.71	0.93	1.17	1.45	1.75	2.08

Erection Sag/Tension Tables for 32 mm² Copper Conductors Basic Span 50 m (Tables 17—19)

Table 17 — Bare Conductors

Temp °C	Tension (kgf)	Sag (m) for Span Length (m)								
		20	25	30	35	40	45	50	55	60
-5.0	336	0.04	0.07	0.10	0.14	0.18	0.22	0.28	0.33	0.40
0.0	307	0.05	0.08	0.11	0.15	0.19	0.24	0.30	0.37	0.43
5.0	280	0.05	0.08	0.12	0.16	0.21	0.27	0.33	0.40	0.48
10.0	255	0.06	0.09	0.13	0.18	0.23	0.29	0.36	0.44	0.52
15.0	232	0.06	0.10	0.14	0.20	0.26	0.32	0.40	0.48	0.58
20.0	211	0.07	0.11	0.16	0.22	0.28	0.36	0.44	0.53	0.63
25.0	192	0.08	0.12	0.17	0.24	0.31	0.39	0.48	0.58	0.69
30.0	177	0.08	0.13	0.19	0.26	0.34	0.43	0.53	0.64	0.76
35.0	163	0.09	0.14	0.21	0.28	0.36	0.46	0.57	0.69	0.82
40.0	151	0.10	0.15	0.22	0.30	0.39	0.50	0.61	0.74	0.89
45.0	141	0.11	0.16	0.24	0.32	0.42	0.53	0.66	0.80	0.95
50.0	132	0.11	0.18	0.25	0.34	0.45	0.57	0.70	0.85	1.01

Table 18 — PVC-covered Conductors (Sections)

Temp °C	Tension (kgf)	Sag (m) for Span Length (m)								
		20	25	30	35	40	45	50	55	60
-5.0	310	0.06	0.09	0.13	0.17	0.23	0.29	0.35	0.43	0.51
0.0	285	0.06	0.10	0.14	0.19	0.25	0.31	0.38	0.46	0.55
5.0	261	0.07	0.10	0.15	0.21	0.27	0.34	0.42	0.51	0.60
10.0	240	0.07	0.11	0.16	0.22	0.29	0.37	0.46	0.55	0.66
15.0	221	0.08	0.12	0.18	0.24	0.32	0.40	0.49	0.60	0.71
20.0	204	0.09	0.13	0.19	0.26	0.34	0.43	0.54	0.65	0.77
25.0	189	0.09	0.14	0.21	0.28	0.37	0.47	0.58	0.70	0.83
30.0	177	0.10	0.15	0.22	0.30	0.40	0.50	0.62	0.75	0.89
35.0	165	0.11	0.17	0.24	0.32	0.42	0.54	0.66	0.80	0.95
40.0	155	0.11	0.18	0.25	0.34	0.45	0.57	0.70	0.85	1.01
45.0	147	0.12	0.19	0.27	0.37	0.48	0.60	0.75	0.90	1.07
50.0	139	0.13	0.20	0.28	0.39	0.50	0.64	0.79	0.95	1.13

Table 19 — PVC-covered Conductors (Single Spans)

Temp °C	Tension (kgf)	Sag (m) for Span Length (m)								
		20	25	30	35	40	45	50	55	60
-5.0	336	0.05	0.08	0.12	0.16	0.21	0.26	0.33	0.39	0.47
0.0	307	0.06	0.09	0.13	0.17	0.23	0.29	0.36	0.43	0.51
5.0	280	0.06	0.10	0.14	0.19	0.25	0.32	0.39	0.47	0.56
10.0	255	0.07	0.11	0.15	0.21	0.27	0.35	0.43	0.52	0.62
15.0	232	0.08	0.12	0.17	0.23	0.30	0.38	0.47	0.57	0.68
20.0	211	0.08	0.13	0.19	0.25	0.33	0.42	0.52	0.63	0.75
25.0	192	0.09	0.14	0.20	0.28	0.36	0.46	0.57	0.69	0.82
30.0	177	0.10	0.15	0.22	0.30	0.40	0.50	0.62	0.75	0.89
35.0	163	0.11	0.17	0.24	0.33	0.43	0.54	0.67	0.81	0.97
40.0	151	0.12	0.18	0.26	0.36	0.46	0.59	0.72	0.88	1.04
45.0	141	0.12	0.19	0.28	0.38	0.50	0.63	0.78	0.94	1.12
50.0	132	0.13	0.21	0.30	0.41	0.53	0.67	0.83	1.00	1.19

Erection Sag/Tension Tables for 50 mm² Aluminium Conductors Basic Span 50 mm (Tables 20—22)

Table 20 — Bare Conductors

Temp °C	Tension (kgf)	Sag (m) for Span Length (m)								
		20	25	30	35	40	45	50	55	60
-5.0	303	0.02	0.04	0.05	0.07	0.10	0.12	0.15	0.18	0.22
0.0	268	0.03	0.04	0.06	0.08	0.11	0.14	0.17	0.20	0.24
5.0	235	0.03	0.05	0.07	0.09	0.12	0.16	0.19	0.23	0.28
10.0	202	0.04	0.06	0.08	0.11	0.14	0.18	0.22	0.27	0.32
15.0	172	0.04	0.07	0.09	0.13	0.17	0.21	0.26	0.32	0.38
20.0	145	0.05	0.08	0.11	0.15	0.20	0.25	0.31	0.38	0.45
25.0	122	0.06	0.09	0.13	0.18	0.24	0.30	0.37	0.45	0.53
30.0	104	0.07	0.11	0.16	0.21	0.28	0.35	0.44	0.53	0.63
35.0	90	0.08	0.13	0.18	0.25	0.32	0.41	0.51	0.61	0.73
40.0	79	0.09	0.14	0.21	0.28	0.37	0.47	0.58	0.70	0.83
45.0	70	0.10	0.16	0.23	0.32	0.41	0.52	0.64	0.78	0.93
50.0	64	0.11	0.18	0.26	0.35	0.45	0.57	0.71	0.86	1.02

Table 21 — PVC-covered Conductors (Sections)

Temp °C	Tension (kgf)	Sag (m) for Span Length (m)								
		20	25	30	35	40	45	50	55	60
-5.0	274	0.04	0.06	0.08	0.11	0.15	0.18	0.23	0.28	0.33
0.0	242	0.04	0.06	0.09	0.13	0.17	0.21	0.26	0.31	0.37
5.0	212	0.05	0.07	0.11	0.14	0.19	0.24	0.29	0.36	0.42
10.0	185	0.05	0.08	0.12	0.17	0.22	0.27	0.34	0.41	0.49
15.0	161	0.06	0.10	0.14	0.19	0.25	0.31	0.39	0.47	0.56
20.0	140	0.07	0.11	0.16	0.22	0.28	0.36	0.45	0.54	0.64
25.0	124	0.08	0.13	0.18	0.25	0.32	0.41	0.51	0.61	0.73
30.0	110	0.09	0.14	0.20	0.28	0.36	0.46	0.57	0.69	0.82
35.0	99	0.10	0.16	0.23	0.31	0.40	0.51	0.63	0.76	0.91
40.0	90	0.11	0.17	0.25	0.34	0.44	0.56	0.69	0.84	1.00
45.0	83	0.12	0.19	0.27	0.37	0.48	0.61	0.75	0.91	1.09
50.0	77	0.13	0.20	0.29	0.40	0.52	0.66	0.81	0.98	1.17

Table 22 — PVC-covered Conductors (Single Spans)

Temp °C	Tension (kgf)	Sag (m) for Span Length (m)								
		20	25	30	35	40	45	50	55	60
-5.0	303	0.03	0.05	0.07	0.10	0.13	0.17	0.21	0.25	0.30
0.0	268	0.04	0.06	0.08	0.11	0.15	0.19	0.23	0.28	0.34
5.0	235	0.04	0.07	0.10	0.13	0.17	0.22	0.27	0.32	0.38
10.0	202	0.05	0.08	0.11	0.15	0.20	0.25	0.31	0.37	0.44
15.0	172	0.06	0.09	0.13	0.18	0.23	0.29	0.36	0.44	0.52
20.0	145	0.07	0.11	0.15	0.21	0.27	0.35	0.43	0.52	0.62
25.0	122	0.08	0.13	0.18	0.25	0.33	0.41	0.51	0.62	0.74
30.0	104	0.10	0.15	0.22	0.30	0.39	0.49	0.60	0.73	0.87
35.0	90	0.11	0.17	0.25	0.34	0.45	0.56	0.70	0.84	1.00
40.0	79	0.13	0.20	0.29	0.39	0.51	0.64	0.79	0.96	1.14
45.0	70	0.14	0.22	0.32	0.44	0.57	0.72	0.89	1.07	1.28
50.0	64	0.16	0.24	0.35	0.48	0.63	0.79	0.98	1.18	1.41

Erection Sag/Tension for 70 mm² Copper Conductors Basic Span 50 m (Tables 23—25)

Table 23 — Bare Conductors

Temp °C	Tension (kgf)	Sag (m) for Span Length (m)								
		20	25	30	35	40	45	50	55	60
- 5.0	452	0.07	0.11	0.15	0.21	0.27	0.35	0.43	0.52	0.62
0.0	413	0.08	0.12	0.17	0.23	0.30	0.38	0.47	0.57	0.68
5.0	378	0.08	0.13	0.18	0.25	0.33	0.42	0.51	0.62	0.74
10.0	348	0.09	0.14	0.20	0.27	0.36	0.45	0.56	0.67	0.80
15.0	322	0.10	0.15	0.22	0.30	0.39	0.49	0.60	0.73	0.87
20.0	300	0.10	0.16	0.23	0.32	0.41	0.52	0.65	0.78	0.93
25.0	281	0.11	0.17	0.25	0.34	0.44	0.56	0.69	0.84	1.00
30.0	264	0.12	0.18	0.26	0.36	0.47	0.60	0.73	0.89	1.06
35.0	250	0.12	0.19	0.28	0.38	0.50	0.63	0.78	0.94	1.12
40.0	237	0.13	0.20	0.29	0.40	0.52	0.66	0.82	0.99	1.18
45.0	226	0.14	0.21	0.31	0.42	0.55	0.70	0.86	1.04	1.24
50.0	216	0.14	0.22	0.32	0.44	0.58	0.73	0.90	1.09	1.29

Table 24 — PVC-covered Conductors (Sections)

Temp °C	Tension (kgf)	Sag (m) for Span Length (m)								
		20	25	30	35	40	45	50	55	60
- 5.0	427	0.08	0.13	0.18	0.25	0.32	0.41	0.50	0.61	0.73
0.0	394	0.09	0.14	0.20	0.27	0.35	0.44	0.55	0.66	0.79
5.0	365	0.09	0.15	0.21	0.29	0.38	0.48	0.59	0.71	0.85
10.0	340	0.10	0.16	0.23	0.31	0.41	0.51	0.63	0.77	0.91
15.0	319	0.11	0.17	0.24	0.33	0.43	0.55	0.68	0.82	0.97
20.0	300	0.12	0.18	0.26	0.35	0.46	0.58	0.72	0.87	1.04
25.0	283	0.12	0.19	0.27	0.37	0.49	0.62	0.76	0.92	1.10
30.0	269	0.13	0.20	0.29	0.39	0.51	0.65	0.80	0.97	1.15
35.0	256	0.13	0.21	0.30	0.41	0.54	0.68	0.84	1.02	1.21
40.0	245	0.14	0.22	0.32	0.43	0.56	0.71	0.88	1.07	1.27
45.0	234	0.15	0.23	0.33	0.45	0.59	0.74	0.92	1.11	1.32
50.0	225	0.15	0.24	0.34	0.47	0.61	0.77	0.96	1.16	1.38

Table 25 — PVC-covered Conductors (Single Spans)

Temp °C	Tension (kgf)	Sag (m) for Span Length (m)								
		20	25	30	35	40	45	50	55	60
- 5.0	452	0.08	0.12	0.17	0.23	0.30	0.39	0.48	0.58	0.69
0.0	413	0.08	0.13	0.19	0.26	0.33	0.42	0.52	0.63	0.75
5.0	378	0.09	0.14	0.21	0.28	0.36	0.46	0.57	0.69	0.82
10.0	348	0.10	0.15	0.22	0.30	0.40	0.50	0.62	0.75	0.89
15.0	322	0.11	0.17	0.24	0.33	0.43	0.54	0.67	0.81	0.96
20.0	300	0.12	0.18	0.26	0.35	0.46	0.58	0.72	0.87	1.04
25.0	281	0.12	0.19	0.28	0.38	0.49	0.62	0.77	0.93	1.11
30.0	264	0.13	0.20	0.29	0.40	0.52	0.66	0.82	0.99	1.18
35.0	250	0.14	0.22	0.31	0.42	0.55	0.70	0.86	1.04	1.24
40.0	237	0.15	0.23	0.33	0.45	0.58	0.74	0.91	1.10	1.31
45.0	226	0.15	0.24	0.34	0.47	0.61	0.77	0.95	1.16	1.37
50.0	216	0.16	0.25	0.36	0.49	0.64	0.81	1.00	1.21	1.44

Erection Sag/Tension Tables for 100 mm² Aluminium Conductors Basic Span 50 m (Tables 26—28)

Table 26 — Bare Conductors

Temp °C	Tension (kgf)	Sag (m) for Span Length (m)								
		20	25	30	35	40	45	50	55	60
-5.0	439	0.03	0.05	0.07	0.10	0.13	0.17	0.21	0.25	0.30
0.0	376	0.04	0.06	0.09	0.12	0.15	0.20	0.24	0.29	0.35
5.0	318	0.05	0.07	0.10	0.14	0.18	0.23	0.28	0.34	0.41
10.0	268	0.05	0.08	0.12	0.17	0.22	0.27	0.34	0.41	0.49
15.0	226	0.06	0.10	0.14	0.20	0.26	0.33	0.40	0.49	0.58
20.0	193	0.08	0.12	0.17	0.23	0.30	0.38	0.47	0.57	0.68
25.0	168	0.09	0.13	0.19	0.26	0.34	0.44	0.54	0.65	0.78
30.0	148	0.10	0.15	0.22	0.30	0.39	0.49	0.61	0.74	0.88
35.0	134	0.11	0.17	0.24	0.33	0.43	0.55	0.68	0.82	0.97
40.0	122	0.12	0.19	0.27	0.36	0.47	0.60	0.74	0.90	1.07
45.0	113	0.13	0.20	0.29	0.39	0.51	0.65	0.80	0.97	1.16
50.0	106	0.14	0.21	0.31	0.42	0.55	0.70	0.86	1.04	1.24

Table 27 — PVC-covered Conductors (Sections)

Temp °C	Tension (kgf)	Sag (m) for Span Length (m)								
		20	25	30	35	40	45	50	55	60
-5.0	404	0.04	0.07	0.10	0.14	0.18	0.23	0.28	0.34	0.40
0.0	349	0.05	0.08	0.12	0.16	0.21	0.26	0.32	0.39	0.46
5.0	300	0.06	0.09	0.14	0.18	0.24	0.30	0.38	0.45	0.54
10.0	259	0.07	0.11	0.16	0.21	0.28	0.35	0.43	0.53	0.62
15.0	226	0.08	0.12	0.18	0.24	0.32	0.40	0.50	0.60	0.72
20.0	200	0.09	0.14	0.20	0.28	0.36	0.46	0.56	0.68	0.81
25.0	179	0.10	0.16	0.23	0.31	0.40	0.51	0.63	0.76	0.90
30.0	163	0.11	0.17	0.25	0.34	0.44	0.56	0.69	0.83	0.99
35.0	149	0.12	0.19	0.27	0.37	0.48	0.61	0.75	0.91	1.09
40.0	139	0.13	0.20	0.29	0.40	0.52	0.66	0.81	0.98	1.17
45.0	130	0.14	0.22	0.31	0.43	0.56	0.70	0.87	1.05	1.25
50.0	122	0.15	0.23	0.33	0.45	0.59	0.74	0.92	1.11	1.32

Table 28 — PVC-covered Conductors (Single Spans)

Temp °C	Tension (kgf)	Sag (m) for Span Length (m)								
		20	25	30	35	40	45	50	55	60
-5.0	439	0.04	0.06	0.09	0.13	0.16	0.21	0.26	0.31	0.37
0.0	376	0.05	0.07	0.11	0.15	0.19	0.24	0.30	0.36	0.43
5.0	318	0.06	0.09	0.13	0.17	0.23	0.29	0.35	0.43	0.51
10.0	268	0.07	0.10	0.15	0.21	0.27	0.34	0.42	0.51	0.60
15.0	226	0.08	0.12	0.18	0.24	0.32	0.40	0.50	0.60	0.72
20.0	193	0.09	0.15	0.21	0.29	0.37	0.47	0.58	0.71	0.84
25.0	168	0.11	0.17	0.24	0.33	0.43	0.54	0.67	0.81	0.96
30.0	148	0.12	0.19	0.27	0.37	0.48	0.61	0.76	0.92	1.09
35.0	134	0.13	0.21	0.30	0.41	0.54	0.68	0.84	1.02	1.21
40.0	122	0.15	0.23	0.33	0.45	0.59	0.74	0.92	1.11	1.32
45.0	113	0.16	0.25	0.36	0.49	0.64	0.81	1.00	1.21	1.44
50.0	106	0.17	0.27	0.38	0.52	0.68	0.86	1.07	1.29	1.53

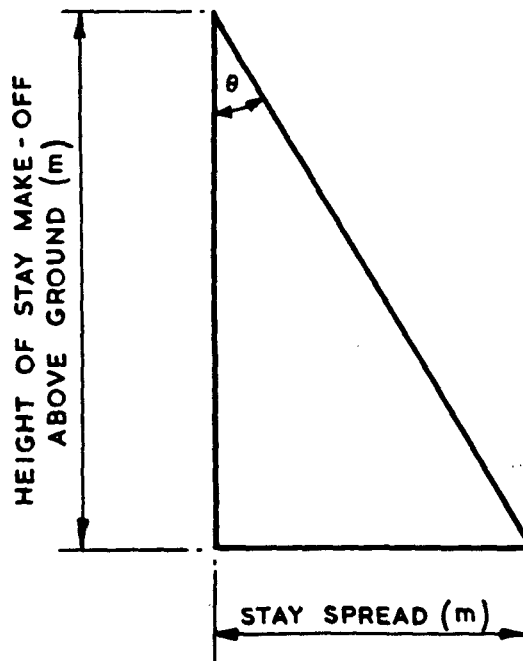
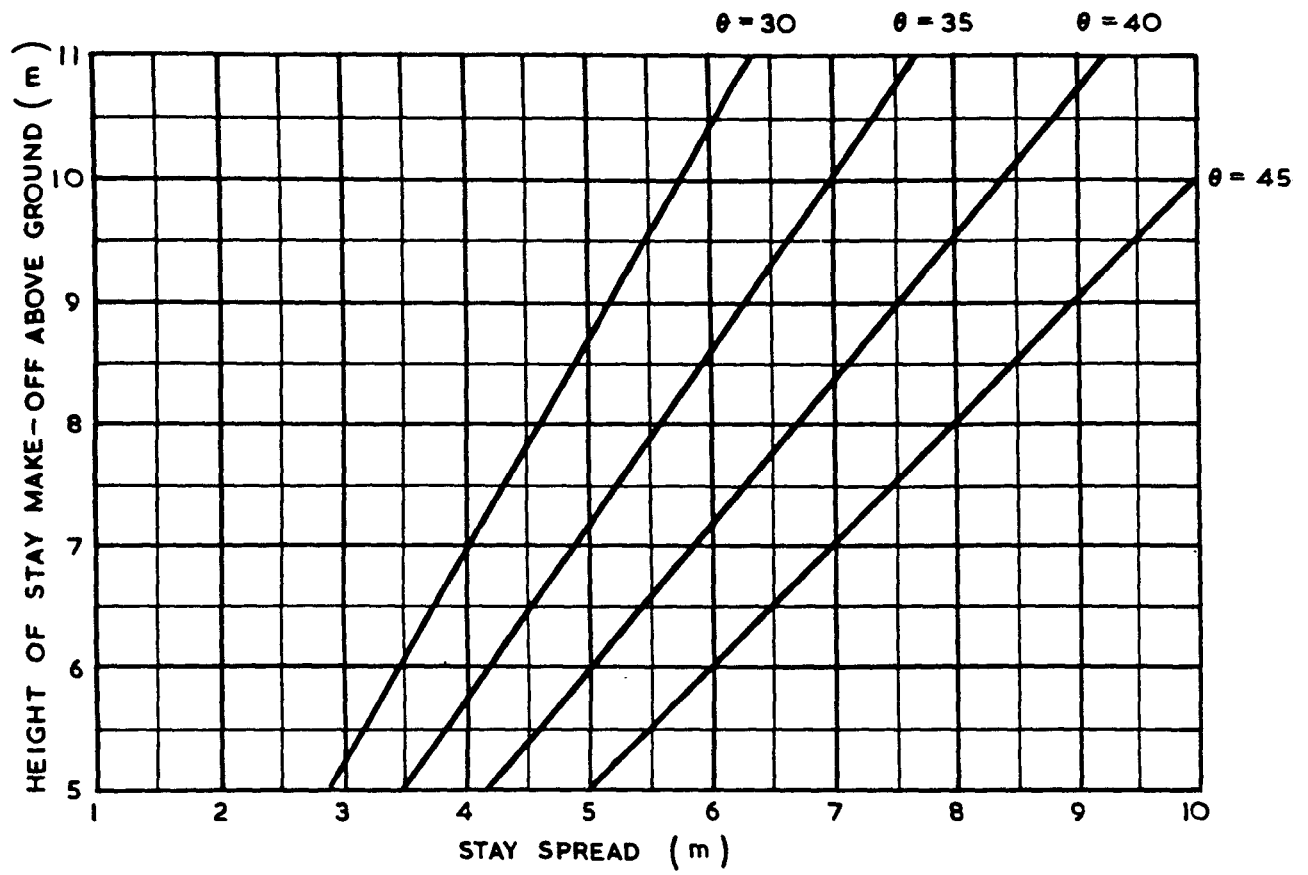


Chart 1 Stay spread

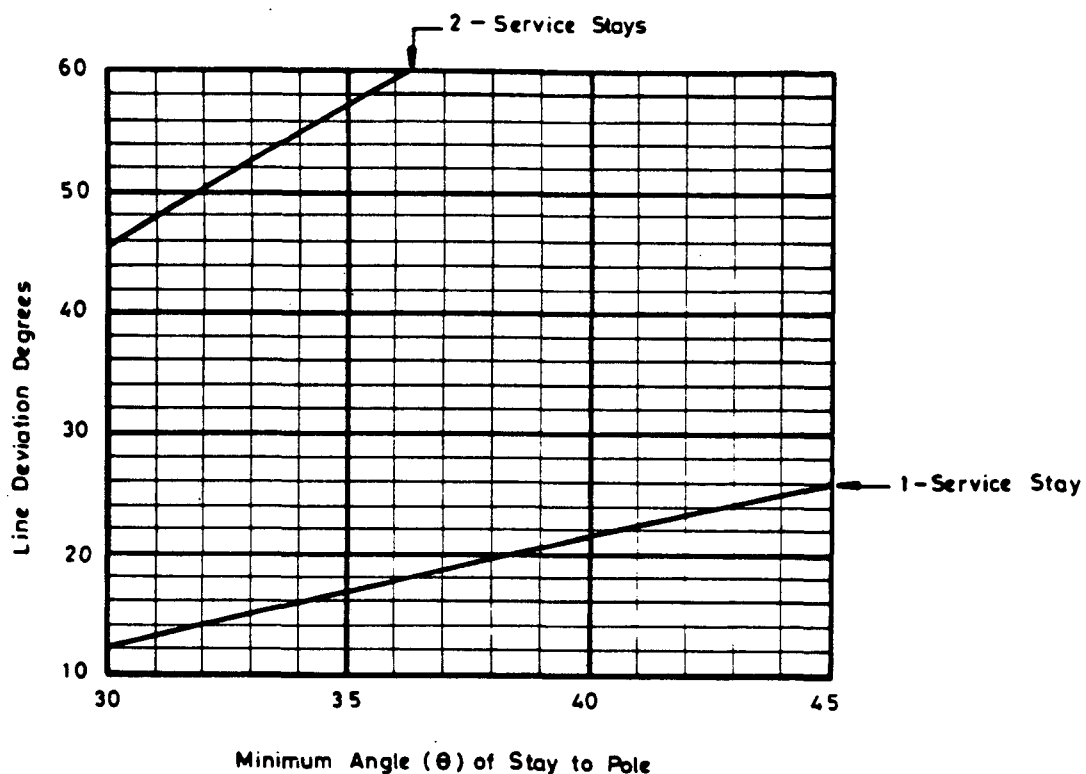


Chart 2 Staying details (service lines) $2 \times 16 \text{ mm}^2$ copper and $2 \times 25 \text{ mm}^2$ aluminium bare and PVC-covered conductors

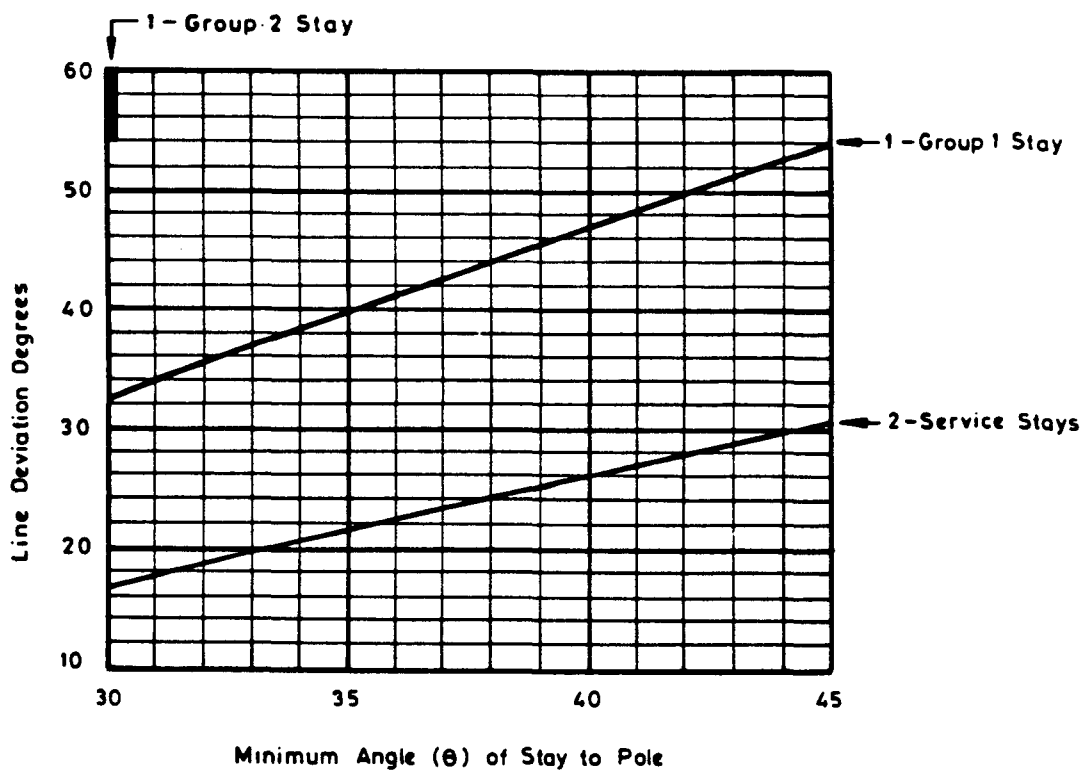


Chart 3 Staying details (service lines) $4 \times 16 \text{ mm}^2$ copper and $4 \times 25 \text{ mm}^2$ aluminium bare and PVC-covered conductors

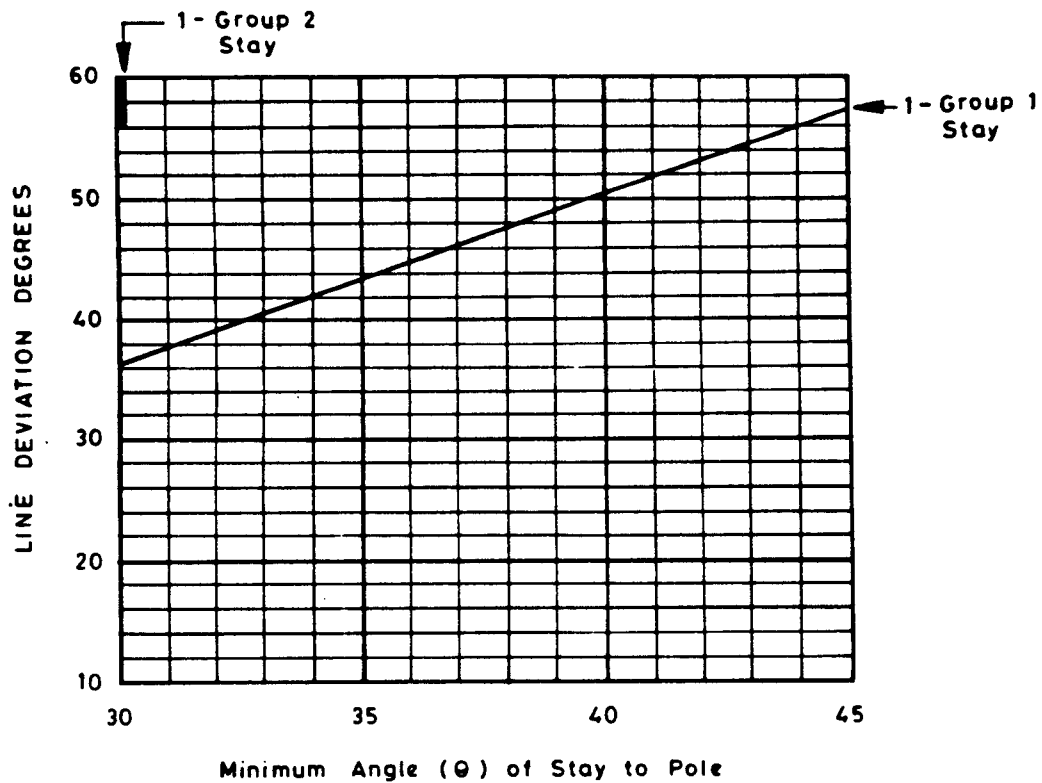


Chart 4 Staying details (distribution lines) $2 \times 32 \text{ mm}^2$ copper and $2 \times 50 \text{ mm}^2$ aluminium bare and PVC-covered conductors

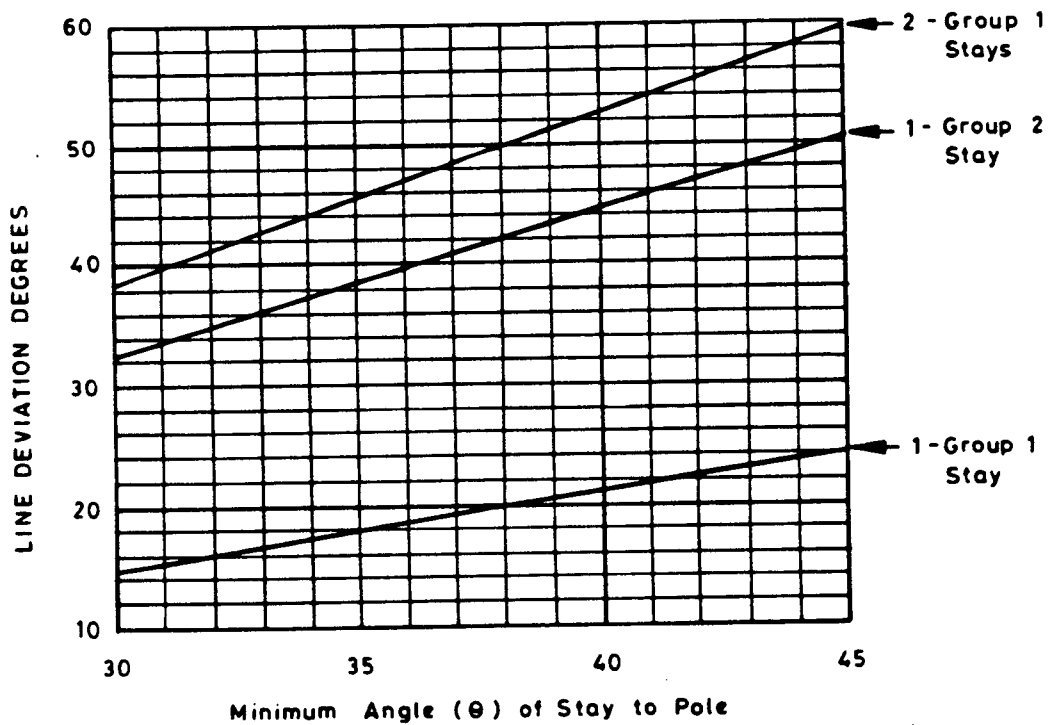


Chart 5 Staying details (distribution lines) $4 \times 32 \text{ mm}^2$ copper and $4 \times 50 \text{ mm}^2$ aluminium bare and PVC-covered conductors

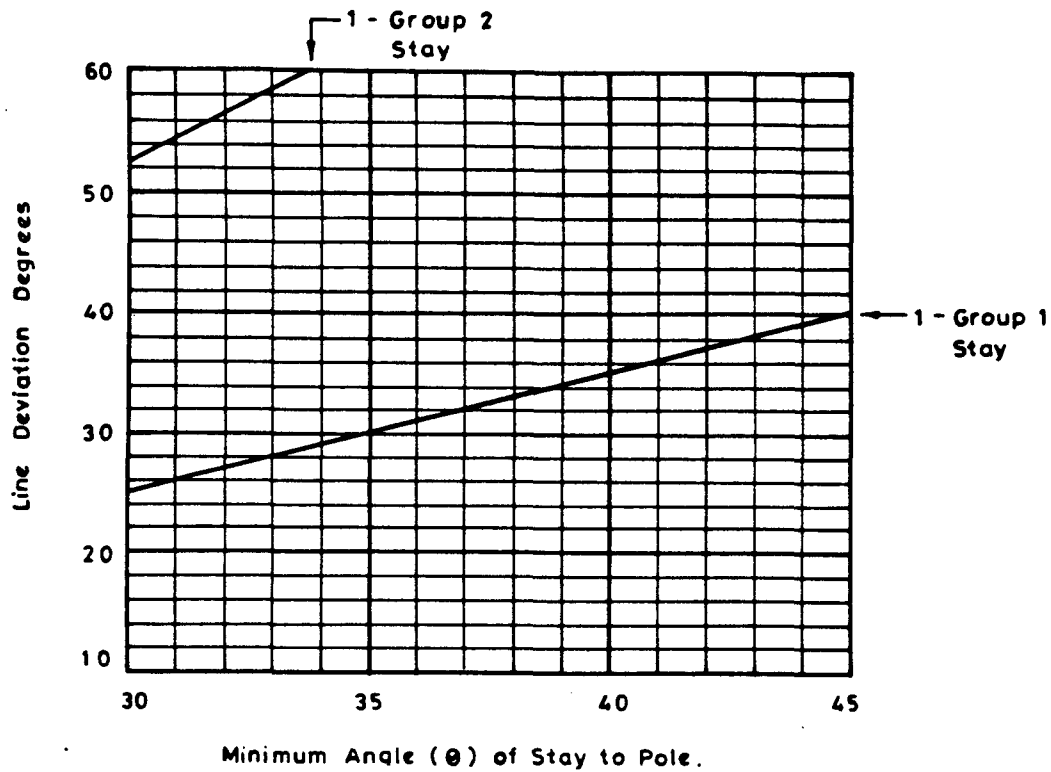


Chart 6 Staying details (distribution lines) $2 \times 70 \text{ mm}^2$ copper and $2 \times 100 \text{ mm}^2$ aluminium bare and PVC-covered conductors

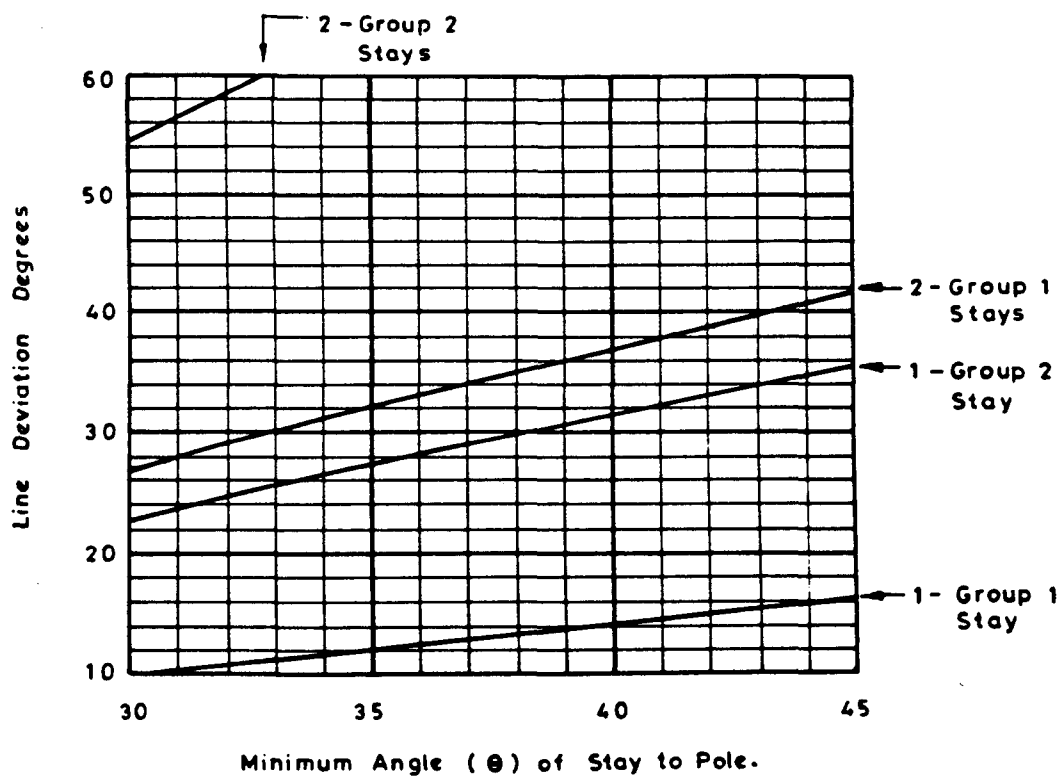


Chart 7 Staying details (distribution lines) $4 \times 70 \text{ mm}^2$ copper and $4 \times 100 \text{ mm}^2$ aluminium bare and PVC-covered conductors

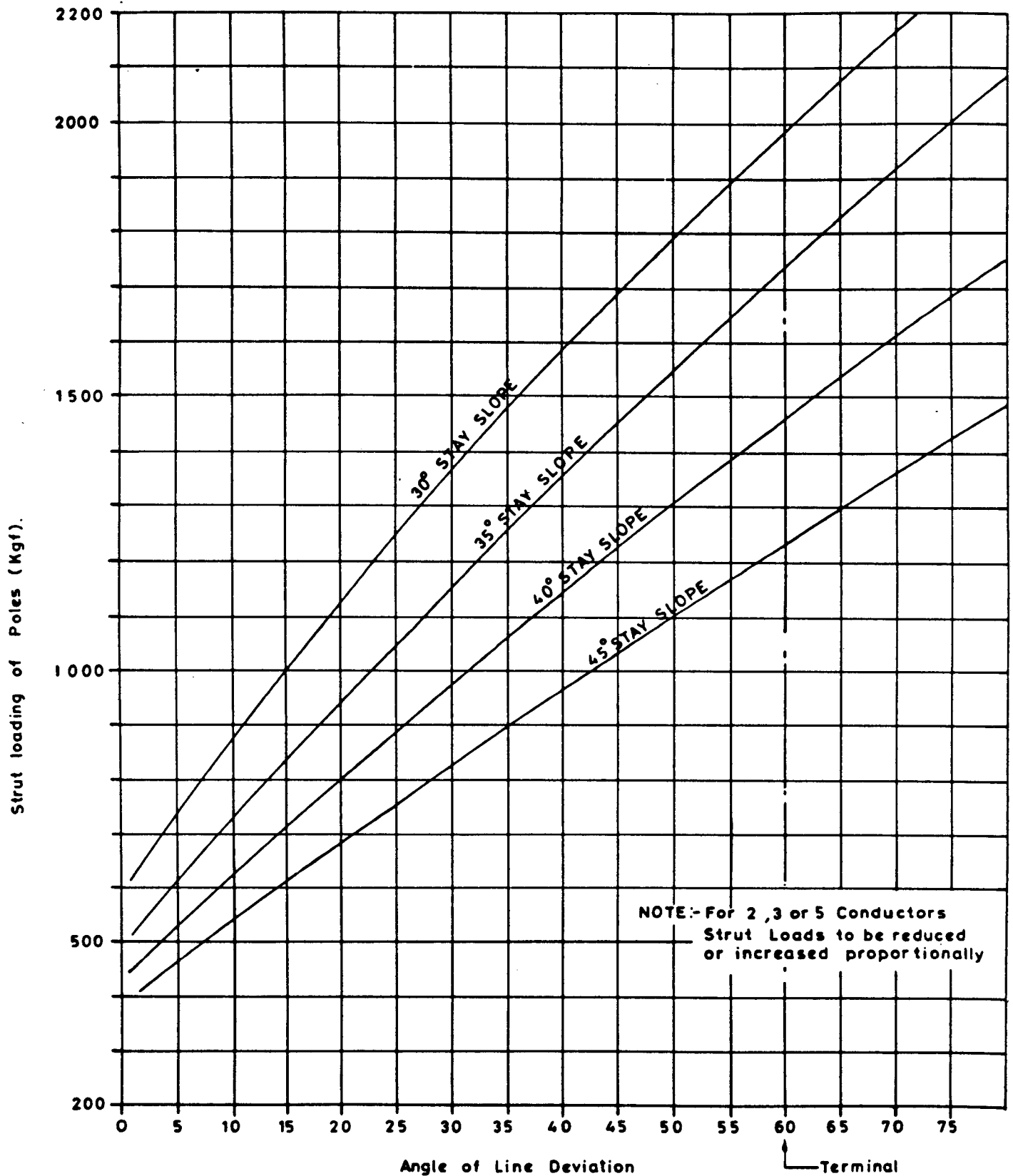


Chart 8 Strut loading of poles 16 mm² copper and 25 mm² aluminium 4 bare and PVC-covered conductors

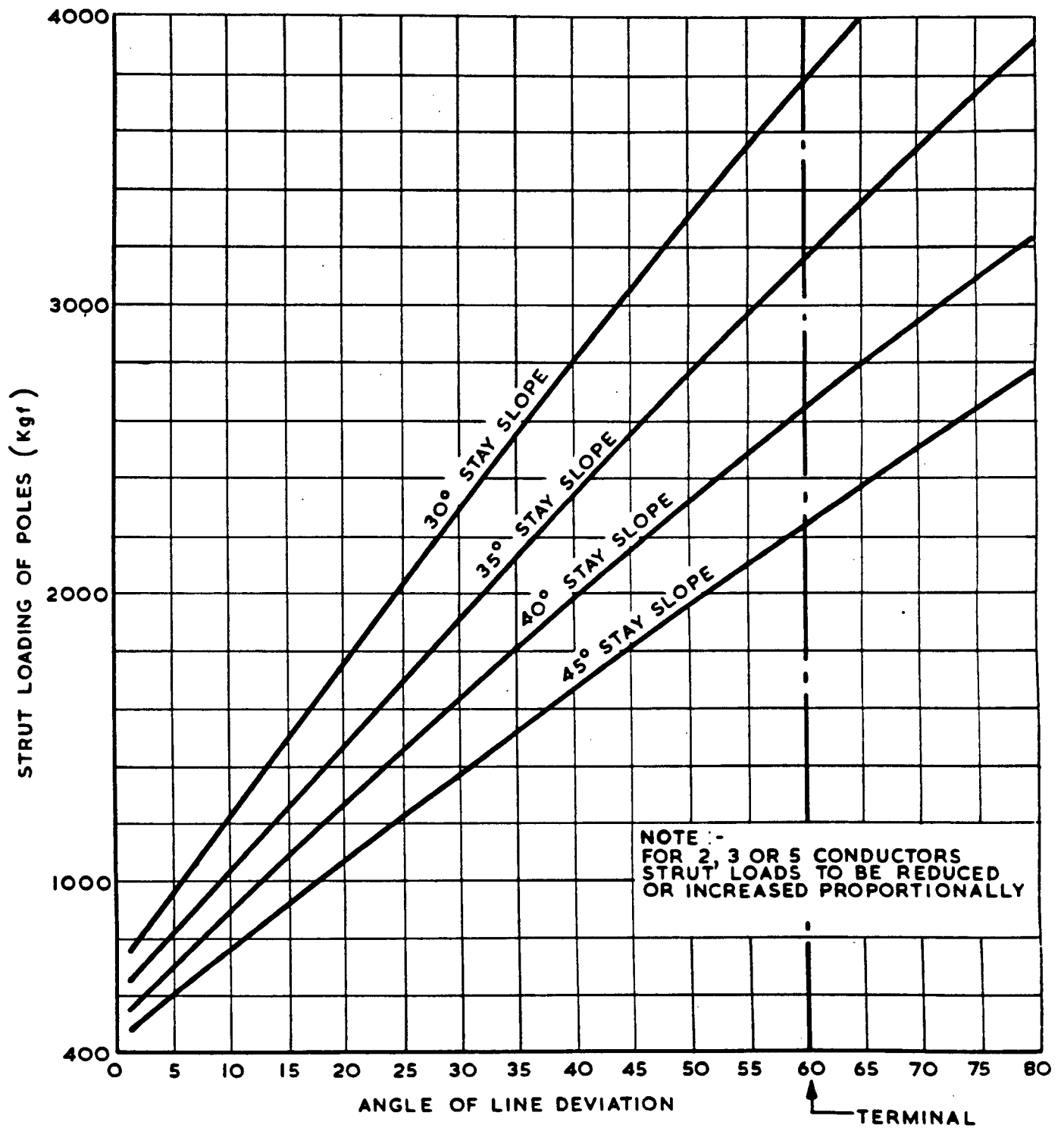


Chart 9 Strut loading of poles 32 mm² copper and 50 mm² aluminium 4 bare or PVC-covered conductors

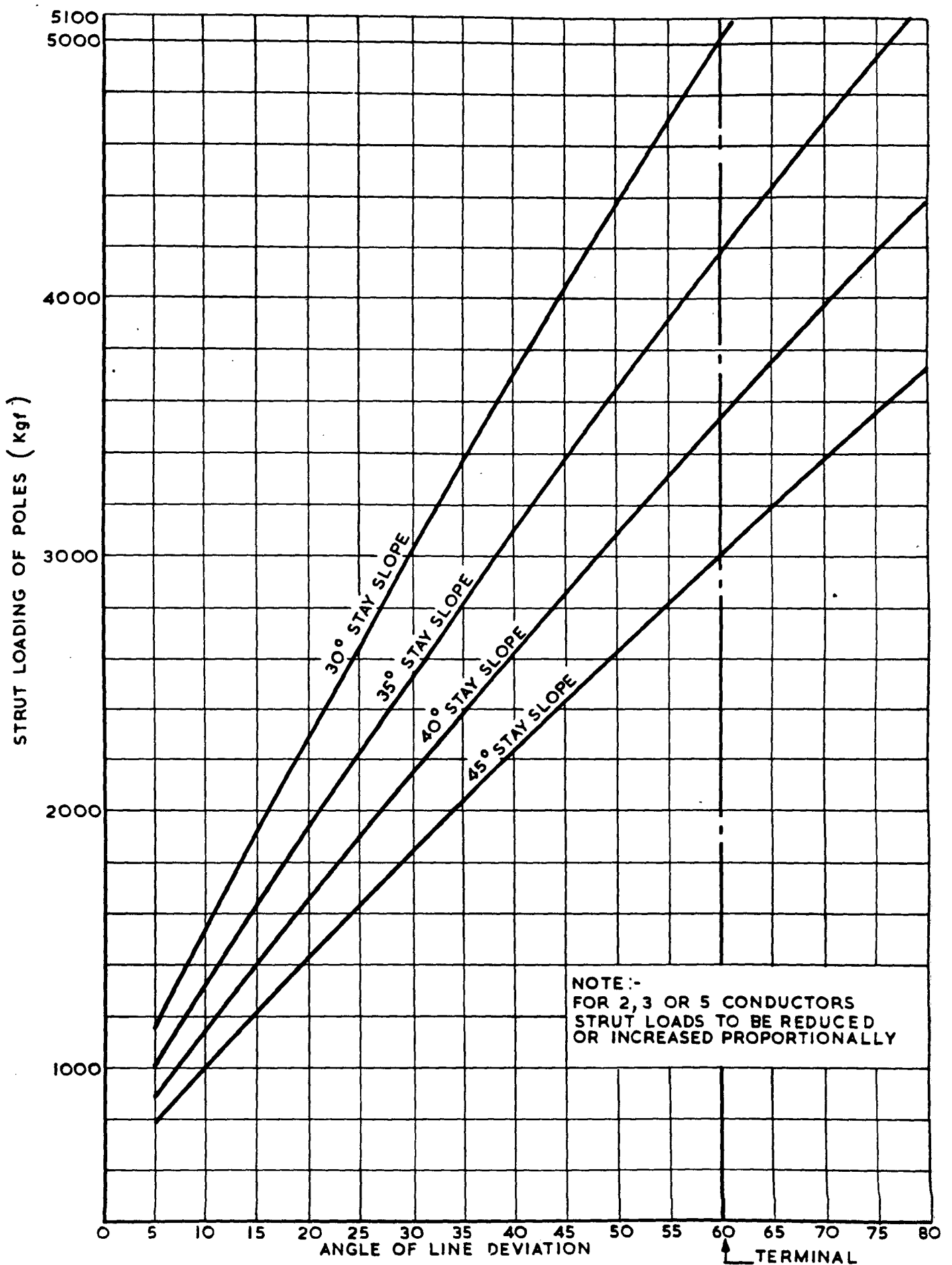
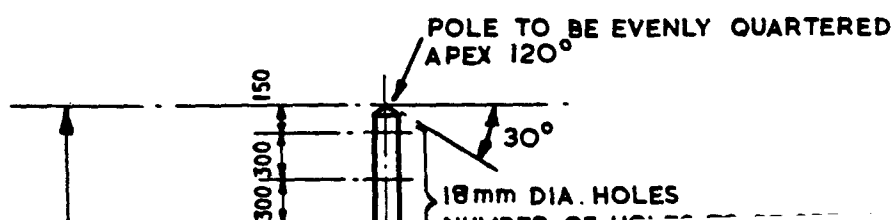


Chart 10 Strut loading of poles 70 mm² copper and 100 mm² aluminium 4 bare or PVC-covered conductors

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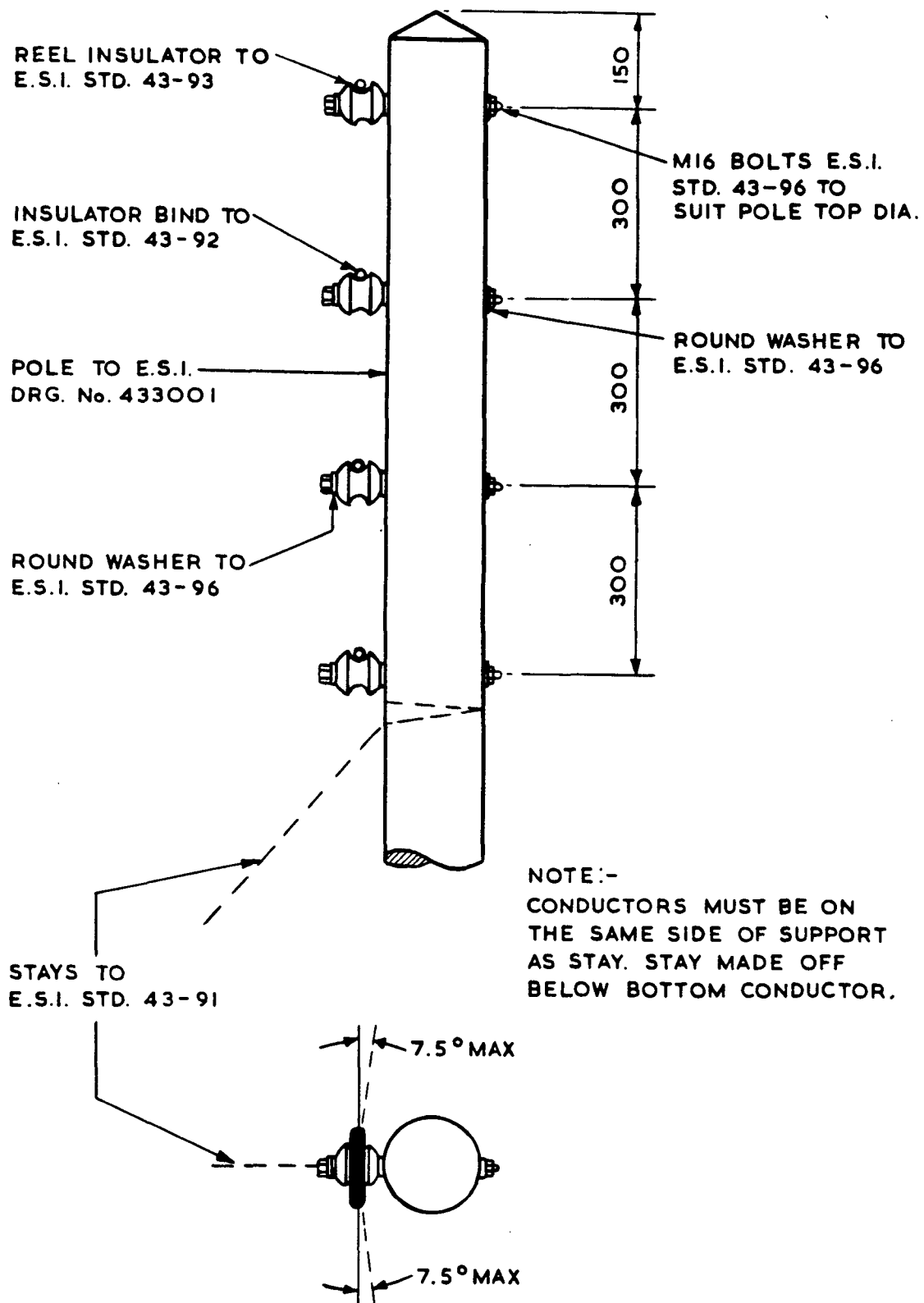


Fig. 2 Intermediate support (deviations up to 15°)
(ESI Drawing 433002)

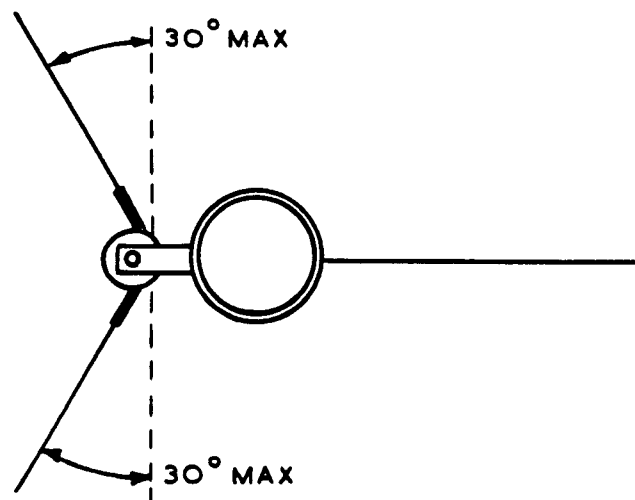
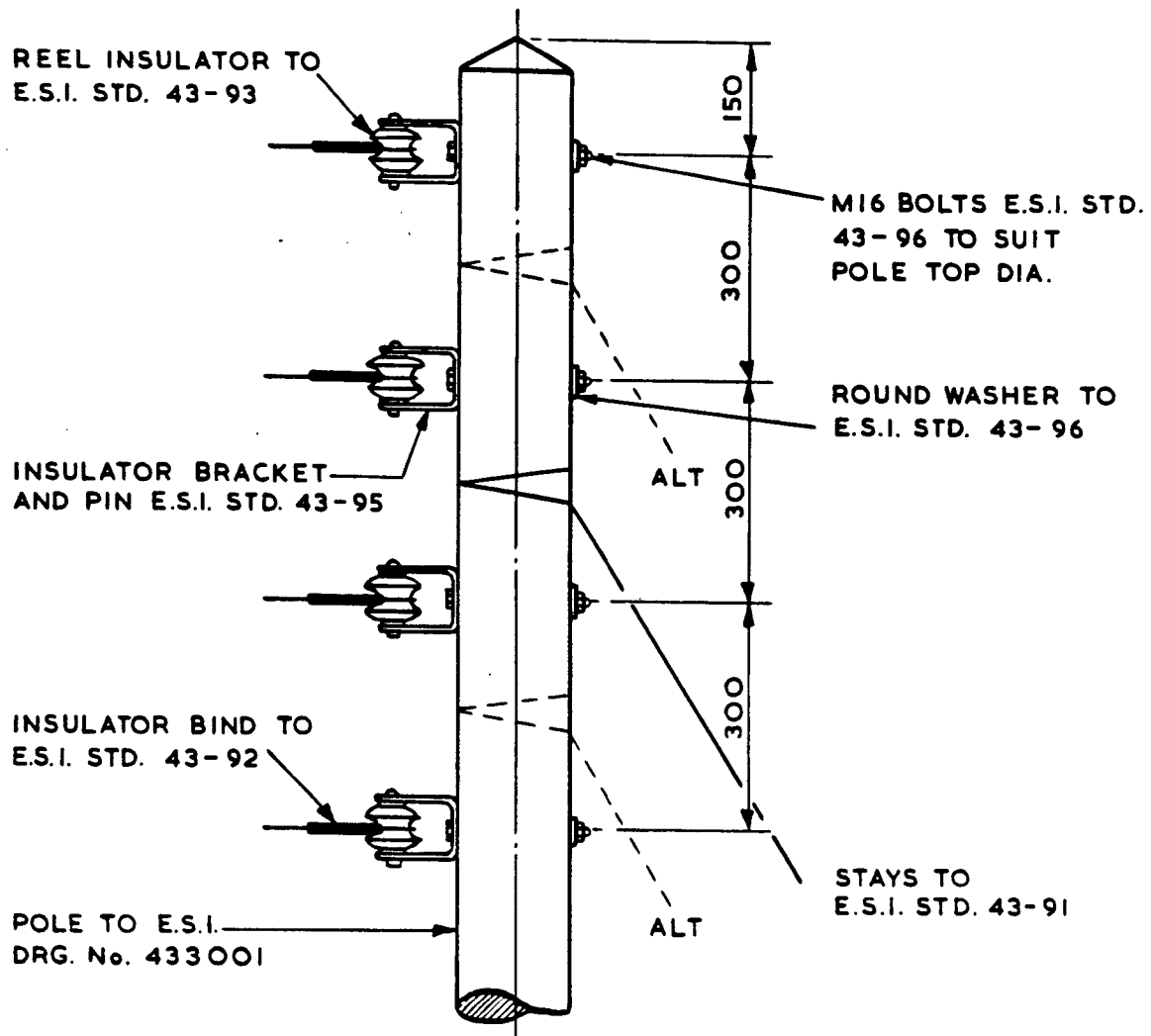


Fig. 3 Intermediate angle support (15° to 60° deviations)
 (ESI Drawing 433003)

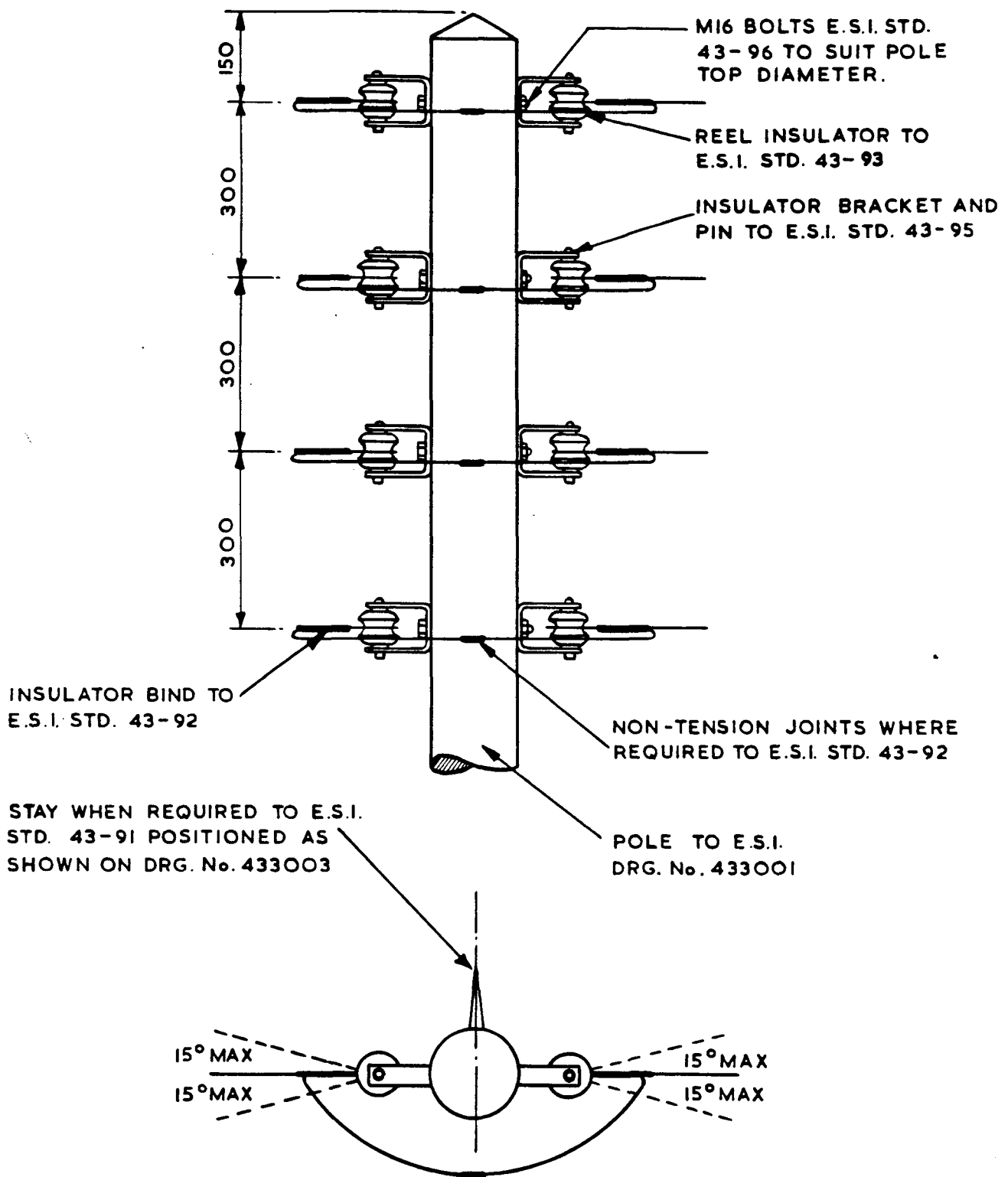


Fig. 4 Section angle support (deviations up to 30°)
(ESI Drawing 433004)

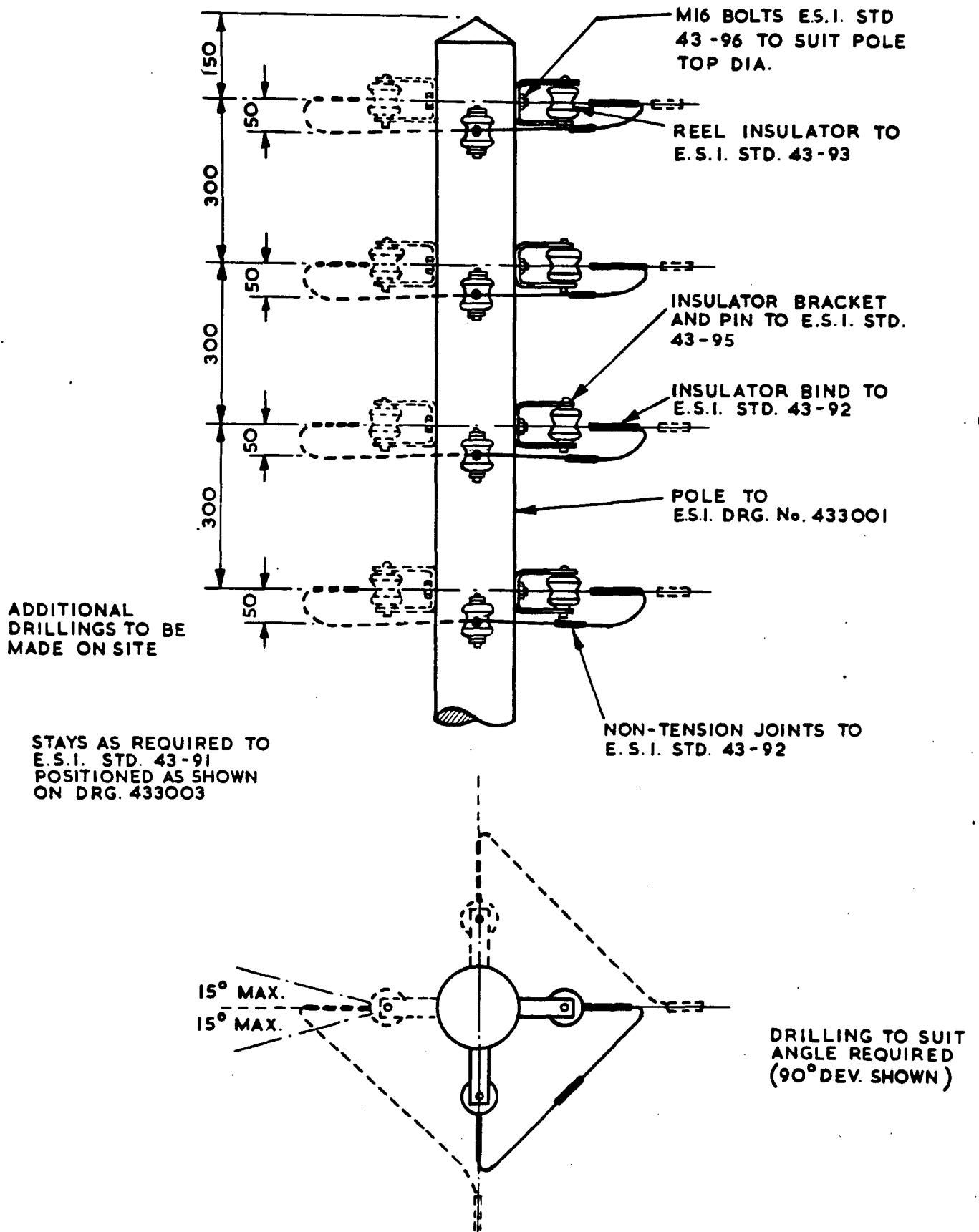


Fig. 5 Section angle support (deviation over 30°) showing three- or four-way arrangement (ESI Drawing 433005)

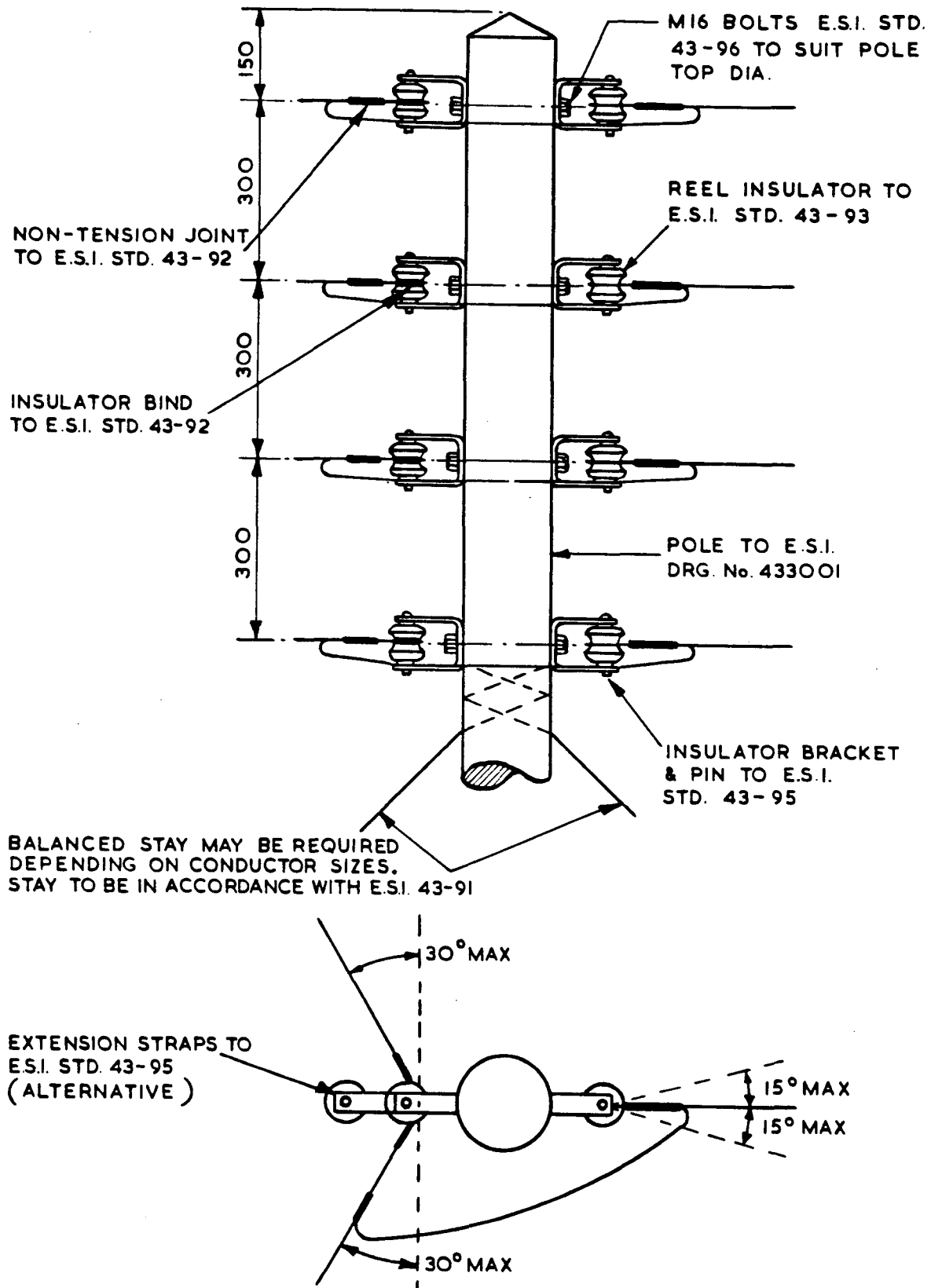


Fig. 6 Intermediate angle support with tee-off connection
(ESI Drawing 433006)

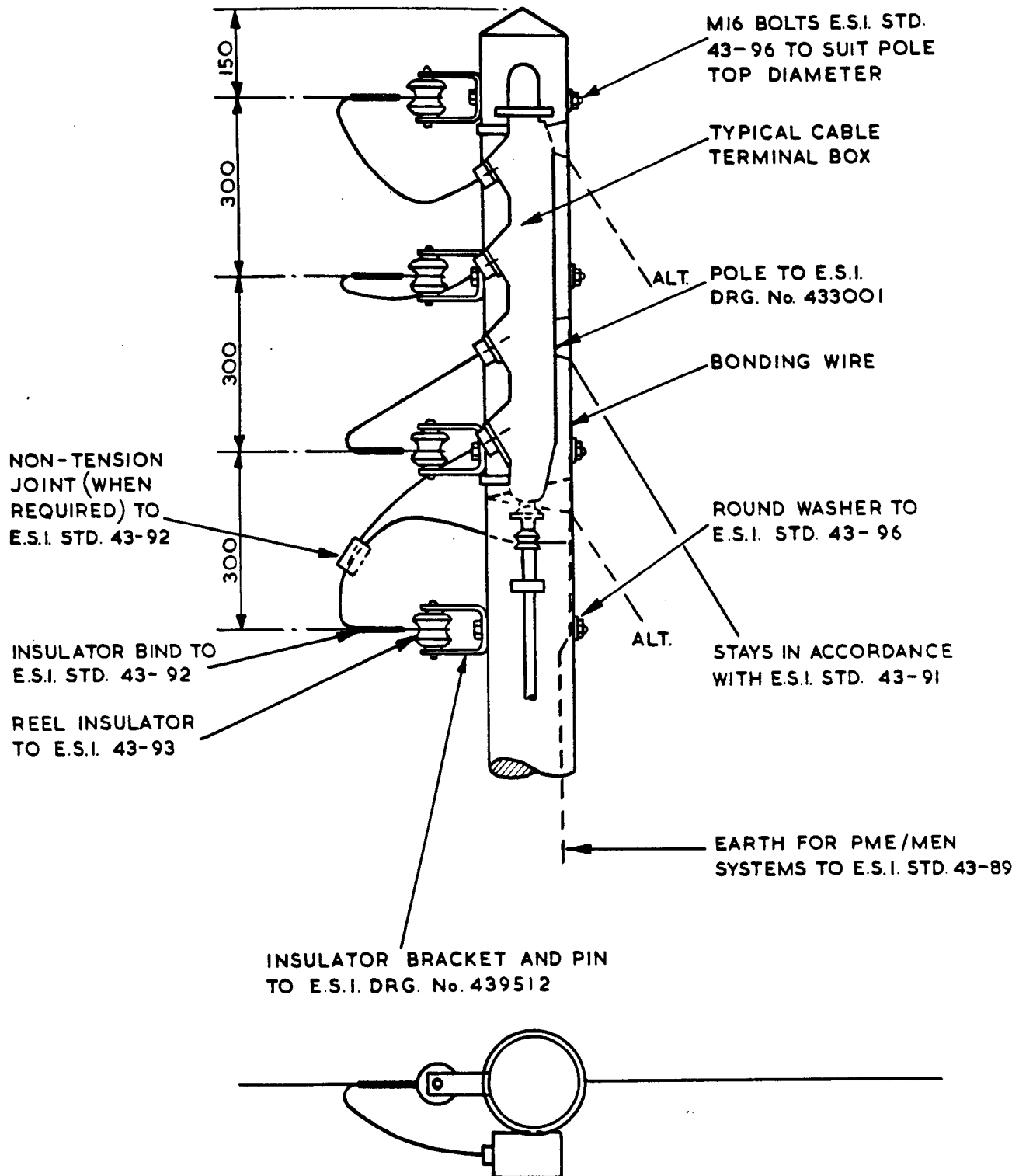


Fig. 7 Terminal support (with cast iron cable box)
(ESI Drawing 433007)

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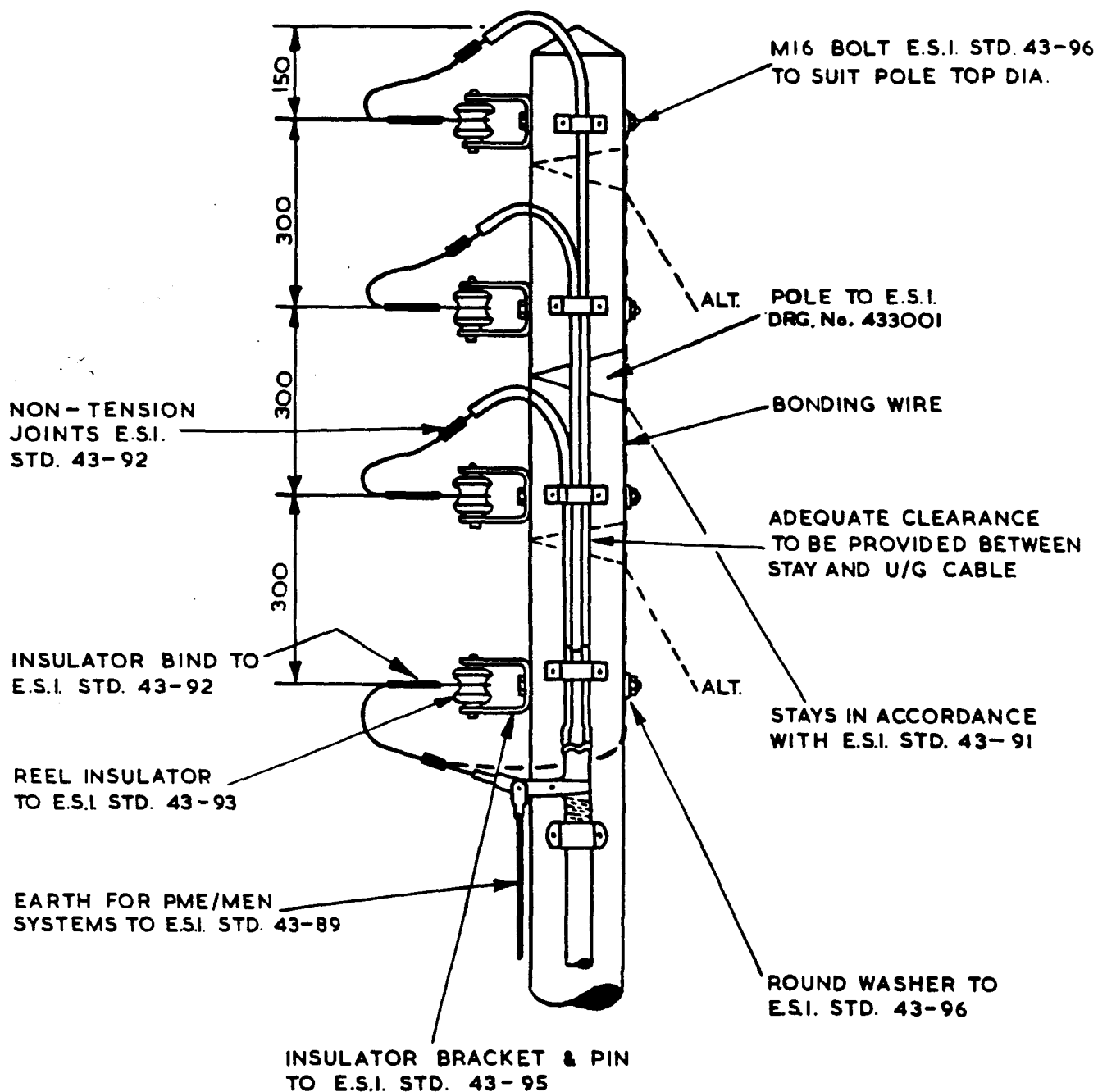


Fig. 8 Terminal support (heat shrinkable termination)
concentric neutral earth cable
(ESI Drawing 433008)

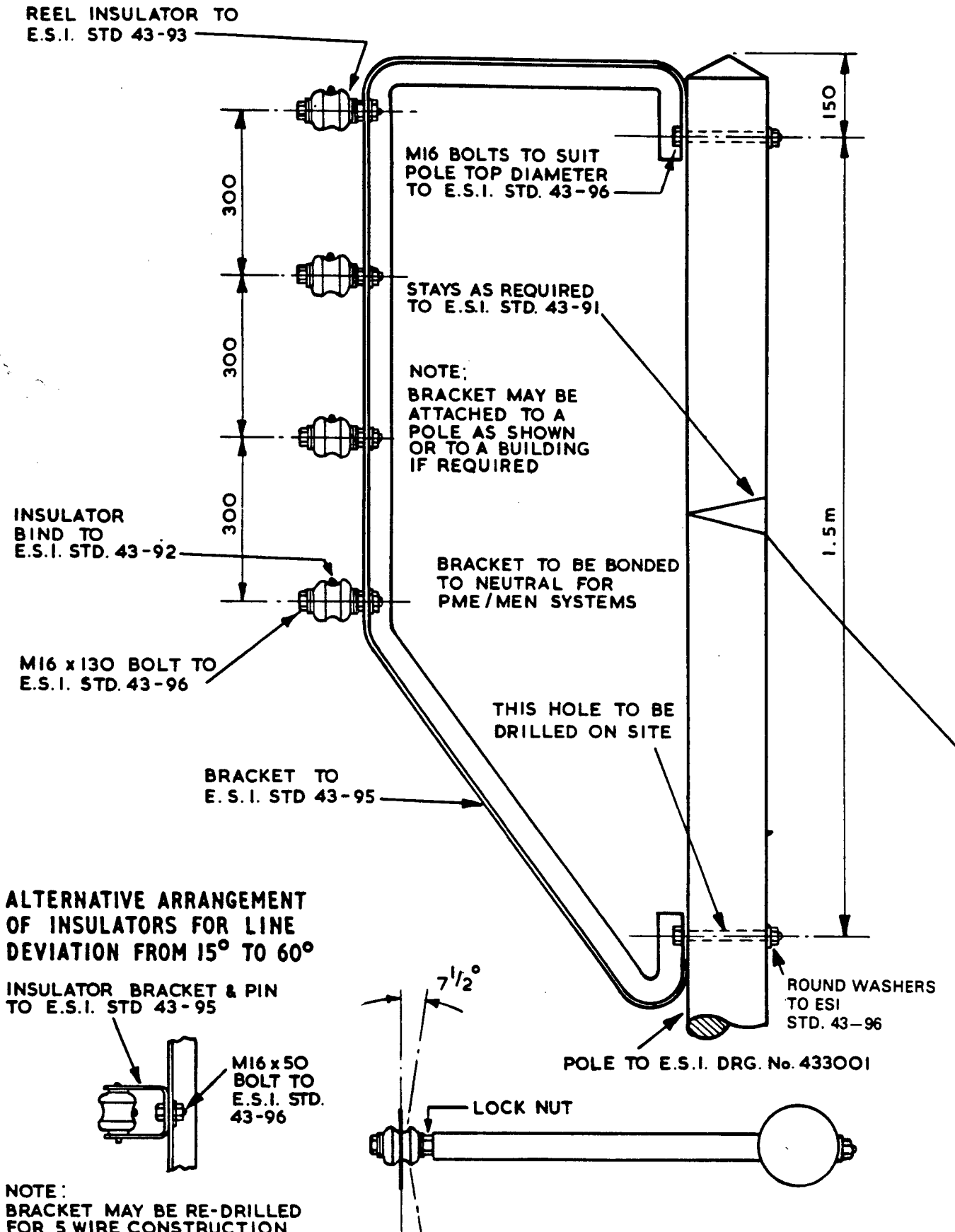


Fig. 9 Intermediate angle support offset arrangement
(ESI Drawing 433009)

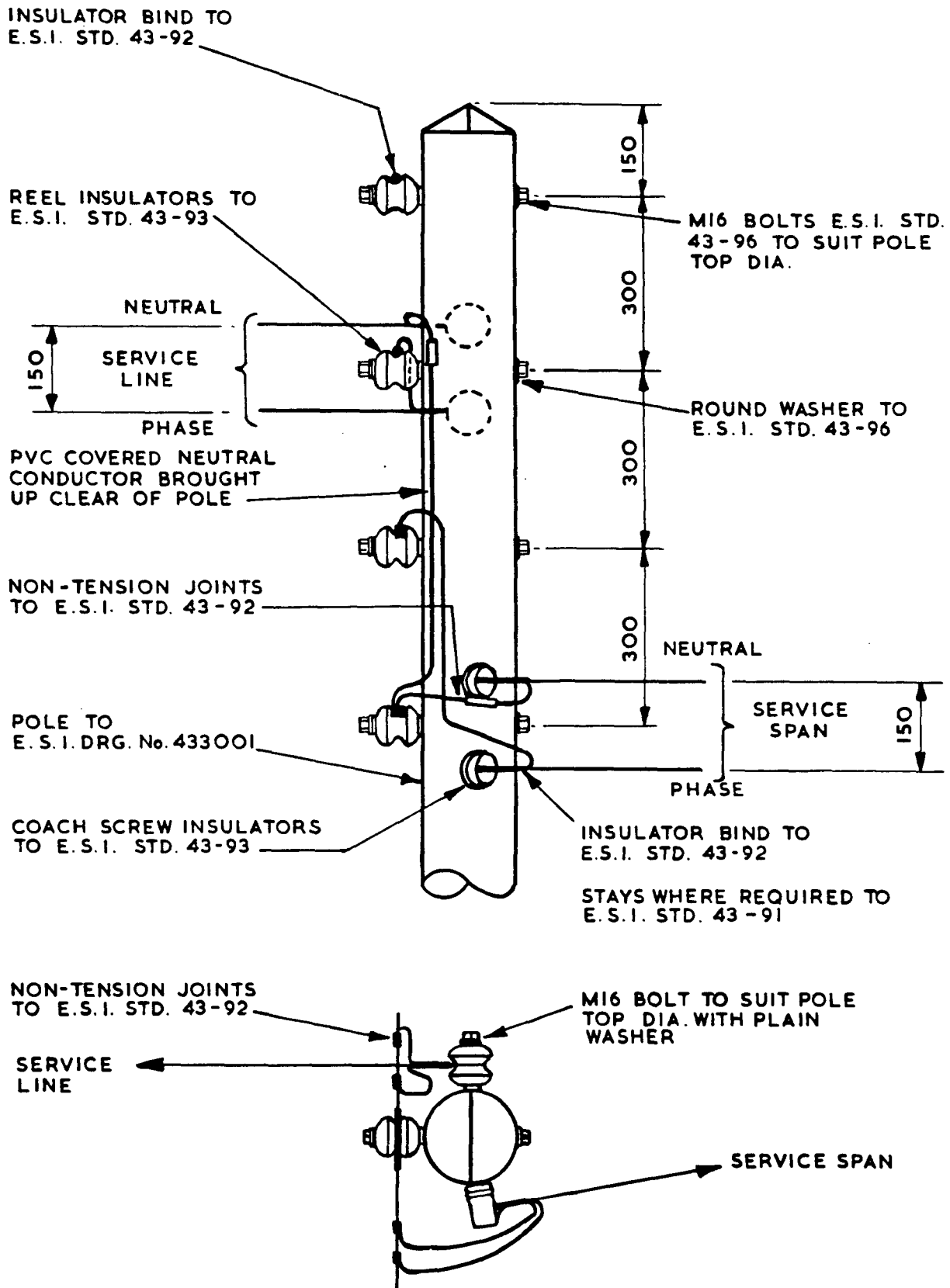
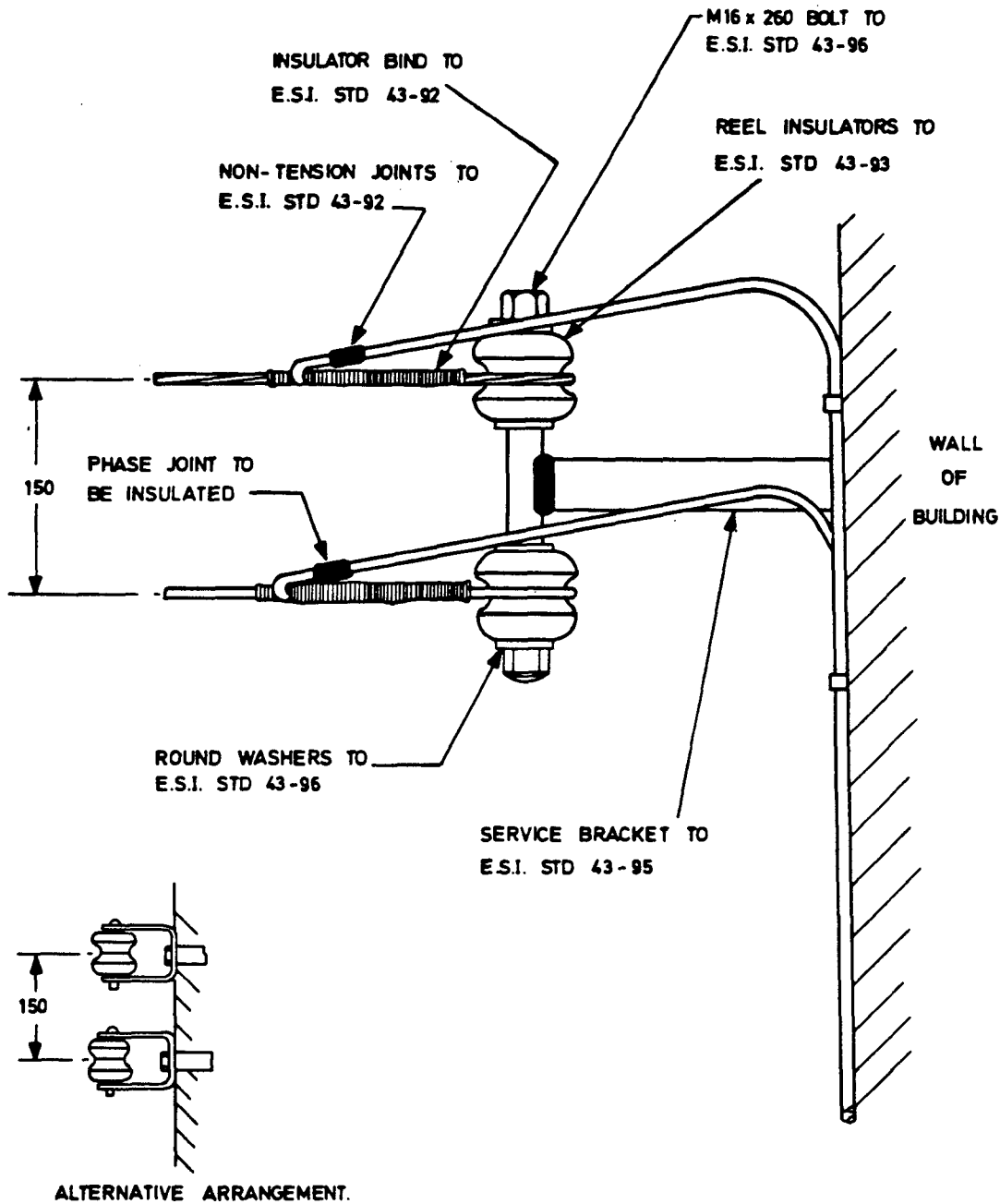
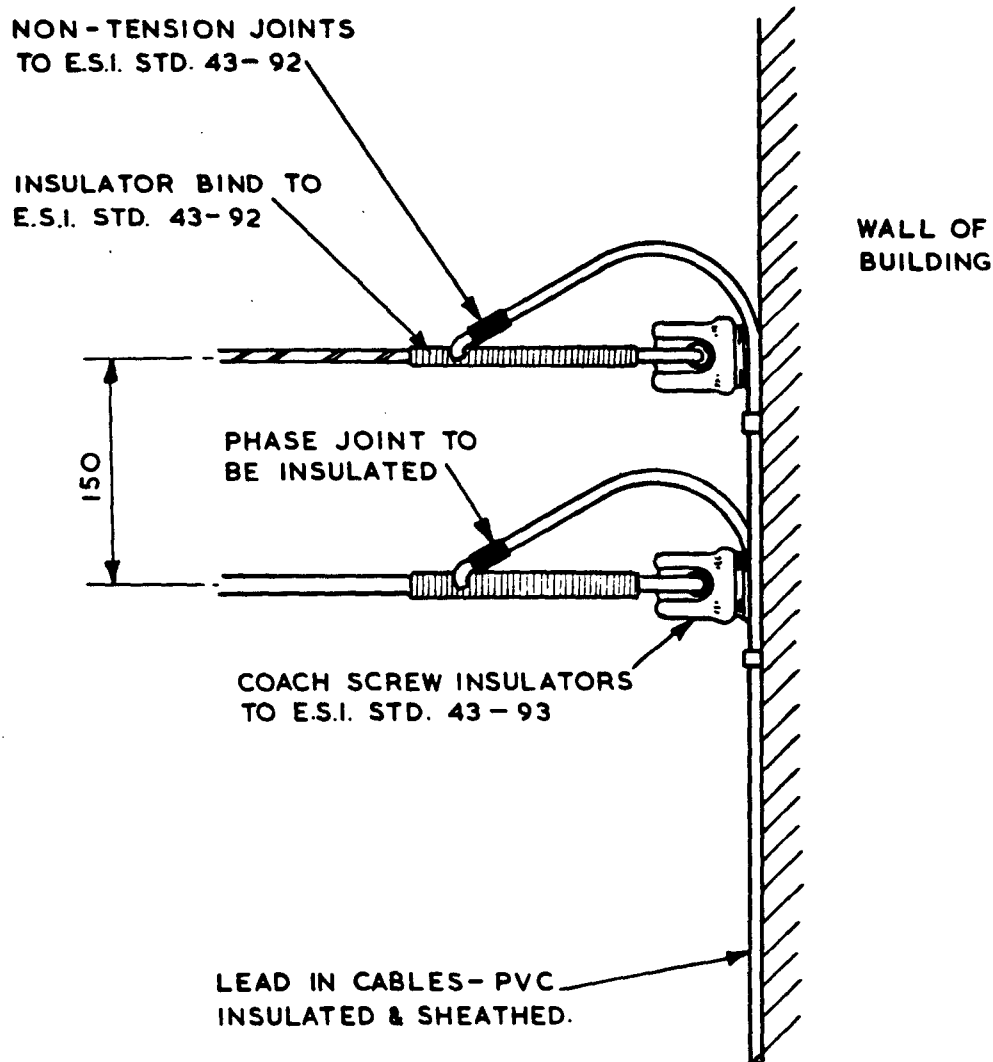


Fig. 10 Typical open-wire service attachment to line support showing alternative insulators (ESI Drawing 433010)



1. WALL FIXINGS TO E.S.I. STD 43-96
2. CABLE CLEATS TO E.S.I. STD 43-98
3. MAX DEVIATION 45° FROM NORMAL.

Fig. 11 Open wire service attachment at building —
service bracket or insulator bracket
(ESI Drawing 433011)



NOTE:-

1. MAXIMUM DEVIATION 45° FROM NORMAL.

Fig. 12 Open wire service attachment to building using coach screw service insulator (ESI Drawing 433012)

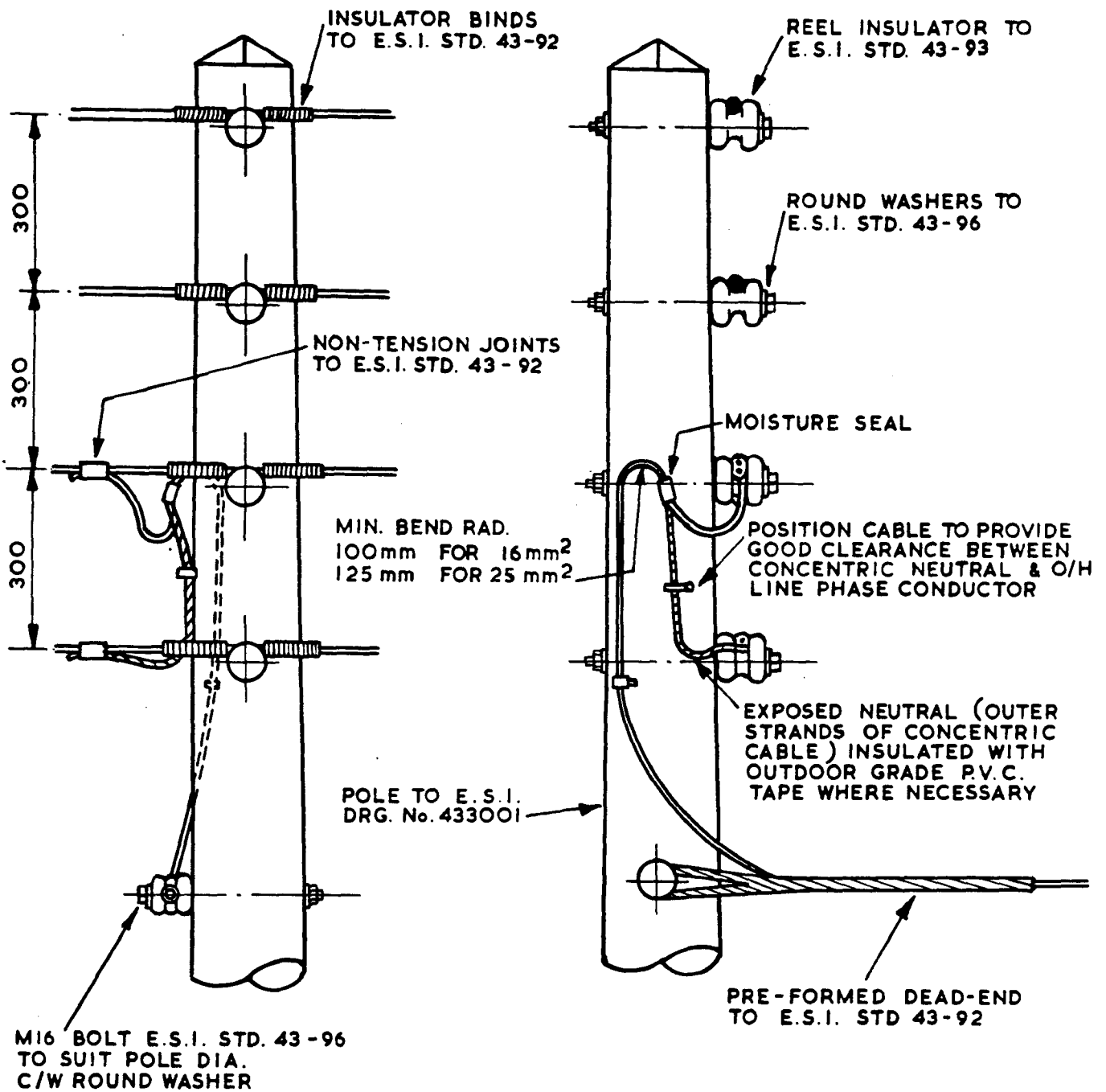
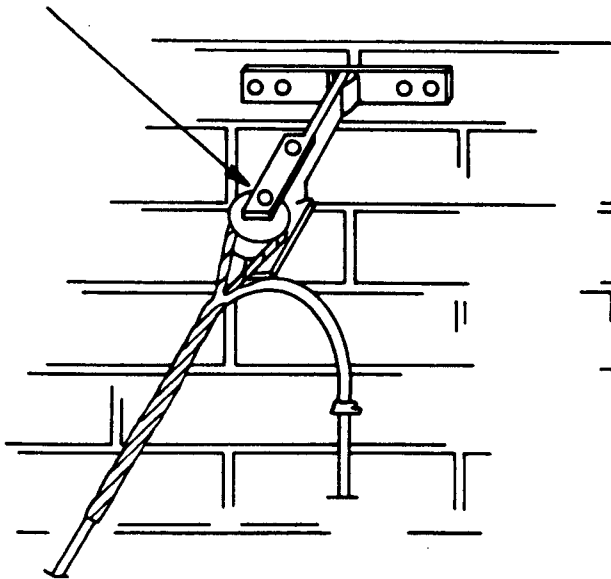
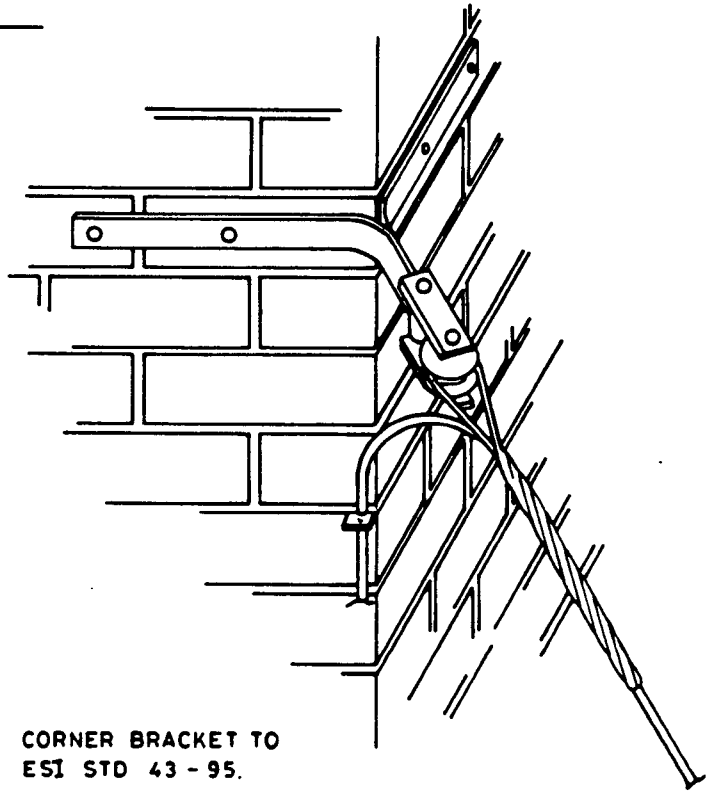


Fig. 13 Typical aerial concentric service cable attachment to line support
 (ESI Drawing 433013)

M16 x 130 BOLTS TO
ESI STD 43-96.

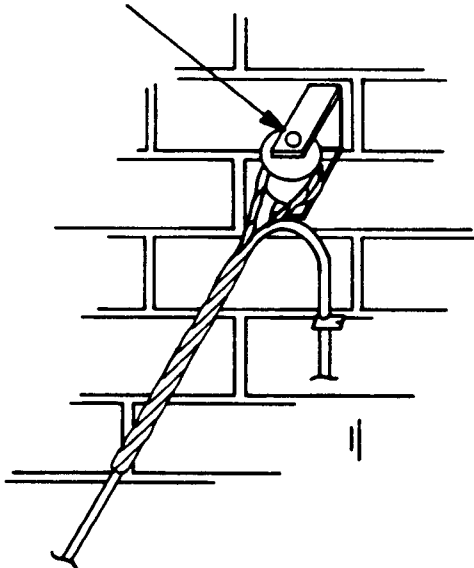


FLAT BRACKET WITH EXTENSION
STRAPS TO ESI STD 43-95.

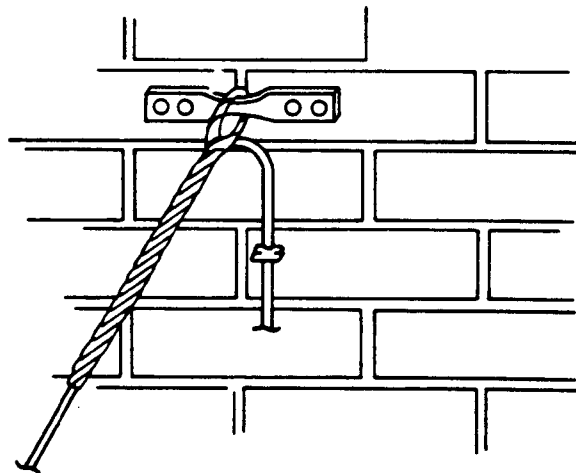


CORNER BRACKET TO
ESI STD 43-95.

M16 x 130 BOLTS TO
ESI STD. 43-96.



INSULATOR BRACKET TO
E.S.I. STD. 43-95.

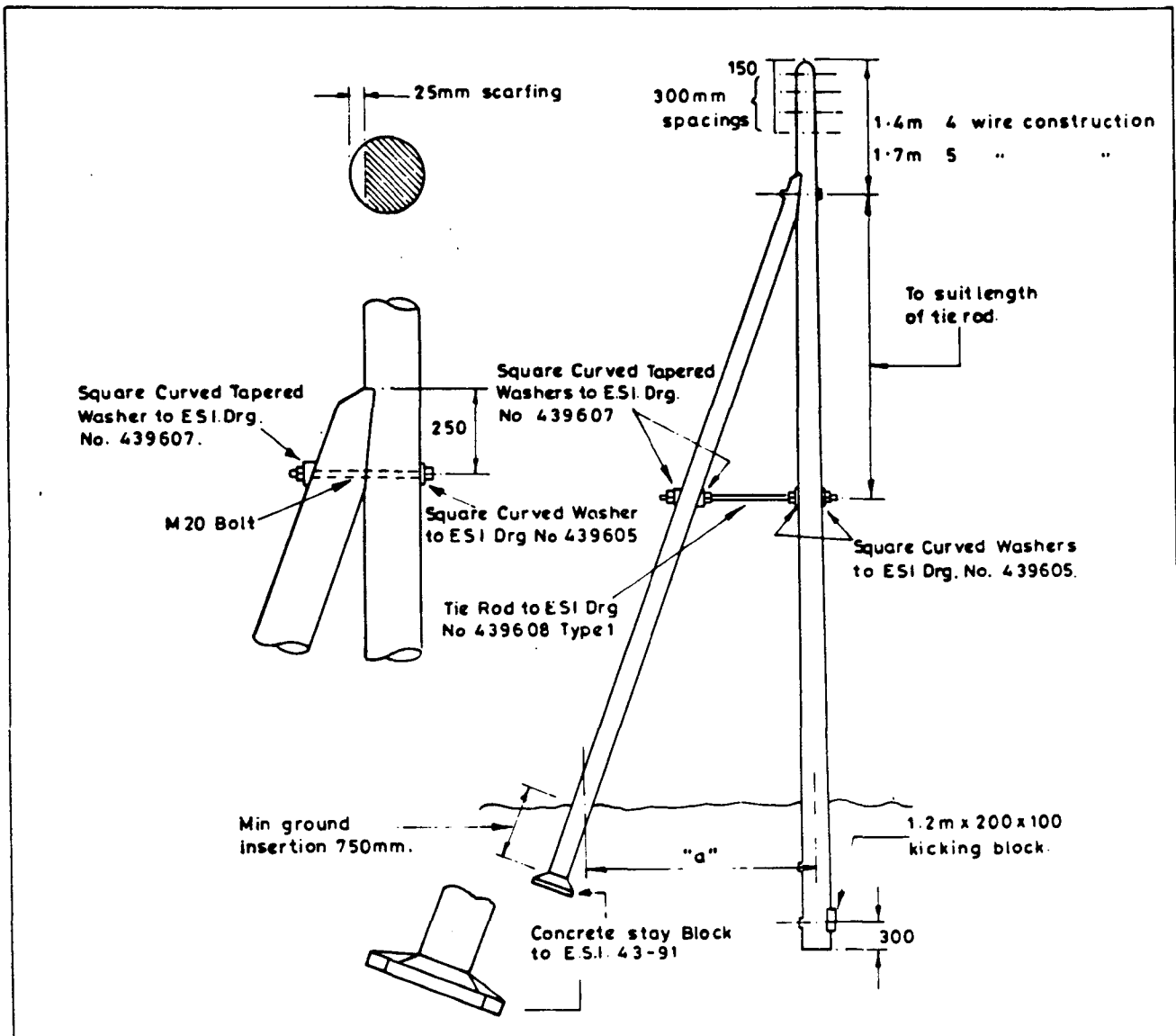


BRACKET TO ESI STD. 43-95.

NOTES:-

1. WALL FIXINGS TO ESI STD. 43-96
2. CABLE CLEATS TO E.S.I. STD. 43-98
3. MAXIMUM DEVIATION 45° FROM
NORMAL.

Fig. 14 Typical aerial concentric service cable attachment to building
(ESI Drawing 433014)



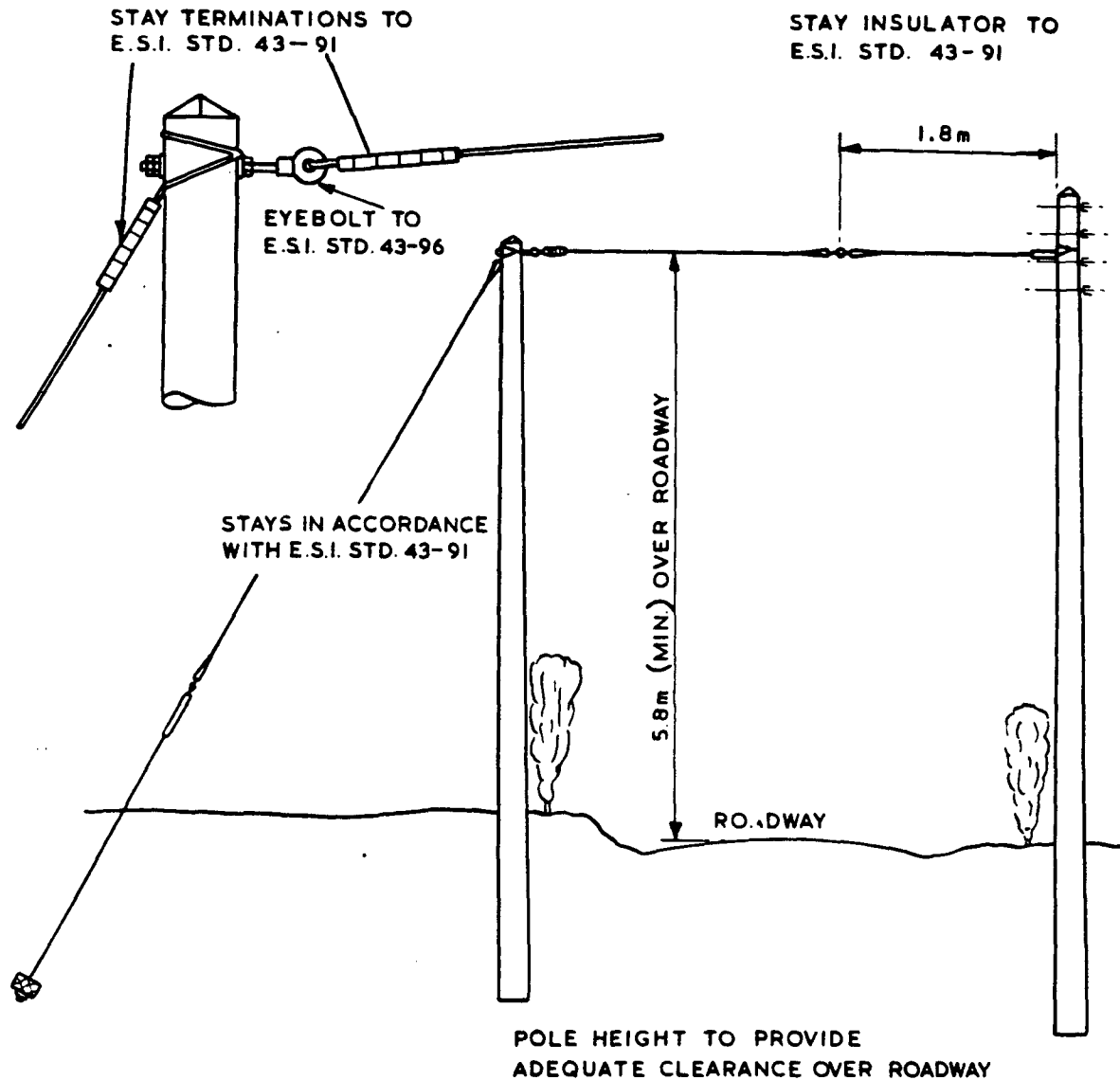
RECOMMENDED SUPPORT & STRUT SIZE

SITE CONDITIONS MUST BE TAKEN INTO CONSIDERATION AND CARE
EXERCISED WHEN SELECTING KICKING AND COMPRESSION BLOCKS

Conductor Type and Size.	Line Deviation	MINIMUM POLE SIZES			MINIMUM STRUT SIZES			Dimension a
		Length(m)	Dia at Pole Top	Dia at 1.5m from Buttend	Length(m)	Dia at Pole Top	Dia at 1.5m from Buttend	
4x16mm ² Cu 4x25mm ² Al	Up to 15°	8.5	150	205	7.0	150	205	2.2m
		9.0	150	205	7.5	150	205	
		9.5	150	210	8.0	150	210	
		10.0	150	210	8.5	150	210	
4x32mm ² Cu 4x50mm ² Al	15° to 30°	8.5	165	230	7.0	165	230	2.3m
		9.0	165	235	7.5	165	235	
		9.5	165	240	8.0	165	240	
		10.0	165	245	8.5	165	245	
4x70mm ² Cu 4x100mm ² Al	Up to 15°	8.5	160	226	7.0	160	225	2.4m
		9.0	160	230	7.5	160	230	
		9.5	160	235	8.0	160	235	
		10.0	160	240	8.5	160	240	
	15° to 30°	8.5	175	240	7.0	175	240	2.5m
		9.0	175	245	7.5	175	245	
		9.5	175	250	8.0	175	250	
		10.0	175	255	8.5	175	255	

All dimensions in mm unless otherwise stated.

Fig. 15 Angle support using strut
(ESI Drawing 433015)



NOTE:-

POLE SIZE AND STAY
IN ACCORDANCE WITH
TABLE 6 & CHARTS 1 TO 10

Fig. 16 Flying stay
(ESI Drawing 433016)

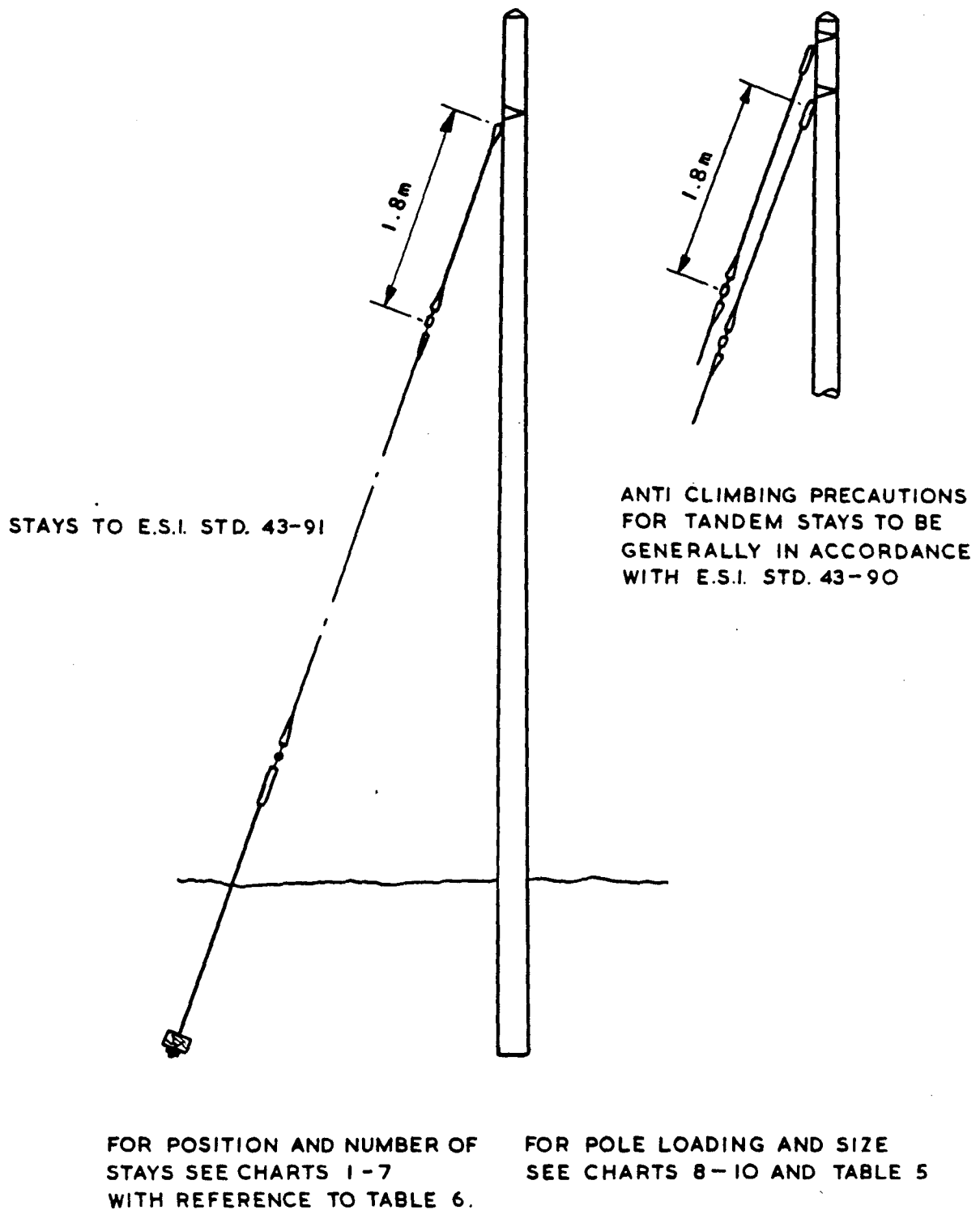


Fig. 17 Typical stay arrangement
 (ESI Drawing 433017)

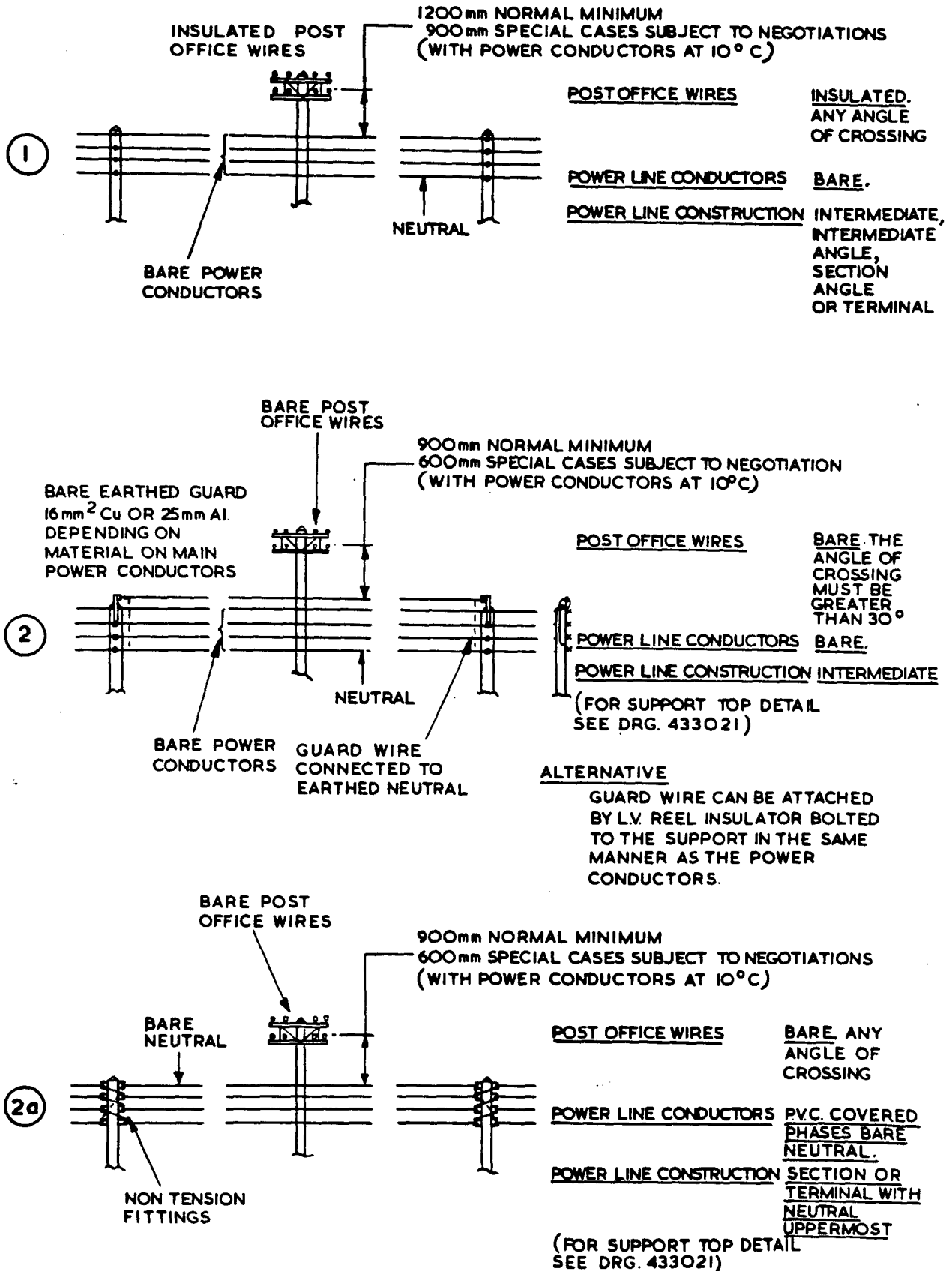
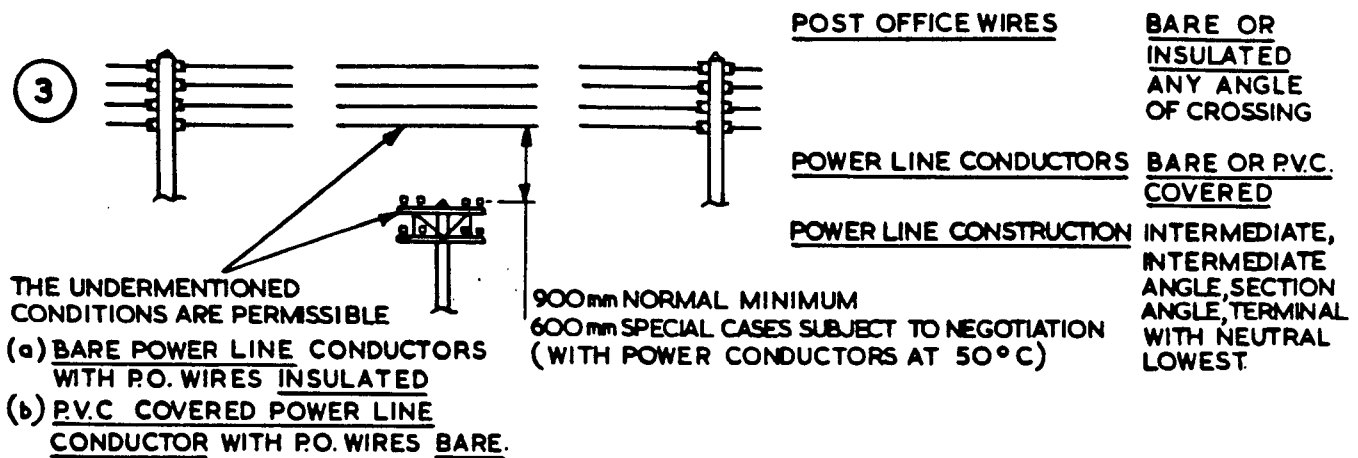


Fig. 19 Power line crossings under Post Office wires
(ESI Drawing 433019)



ANGLE OF CROSSING	METHOD OF PROTECTION TO BE ADOPTED FOR CROSSING		ARRANGEMENT REQUIRED
	POST OFFICE WIRES	POWER CONDUCTORS	
POWER LINE CROSSINGS UNDER POST OFFICE WIRES			
ANY ANGLE	INSULATED	BARE	①
GREATER THAN 30°	BARE	EARTHED GUARD ABOVE BARE POWER LINES.	②
ANY ANGLE	BARE	BARE NEUTRAL ABOVE P.V.C. COVERED PHASE WIRES.	②
POWER LINE CROSSINGS OVER POST OFFICE WIRES			
ANY ANGLE	BARE	BARE NEUTRAL ABOVE P.V.C. COVERED PHASE WIRES.	③
ANY ANGLE	INSULATED	BARE	③

NOTES FOR DRGS. No. 433019 & 433020

- JUMPER LEADS AND OTHER SMALL DETAILS OF THE LINE CONDUCTORS ARE IN PLACES OMITTED FOR CLARITY.
- GUARD TO BE BONDED TO THE NEUTRAL AT EACH END OF SPAN
- TO DETERMINE THE NECESSARY CLEARANCE BETWEEN POWER CONDUCTORS AND POST OFFICE THE FOLLOWING MUST BE OBSERVED:-
 - PROVISION SHOULD BE MADE FOR THE ERECTION OF ADDITIONAL CONDUCTORS
 - IT MUST BE ASCERTAINED FROM THE LOCAL POST OFFICE ENGINEER WHAT SPACE WILL BE REQUIRED, IF ANY, FOR FUTURE DEVELOPMENT OF THE POST OFFICE ROUTE.
- OTHER ARRANGEMENTS ARE PERMISSIBLE INCLUDING POWER CONDUCTORS IN HORIZONTAL FORMATION. FOR DETAILS REFER TO LATEST EDITION OF POST OFFICE MEMORANDUM A80.

Fig. 20 Power line crossings over Post Office wires
(ESI Drwing 433020)

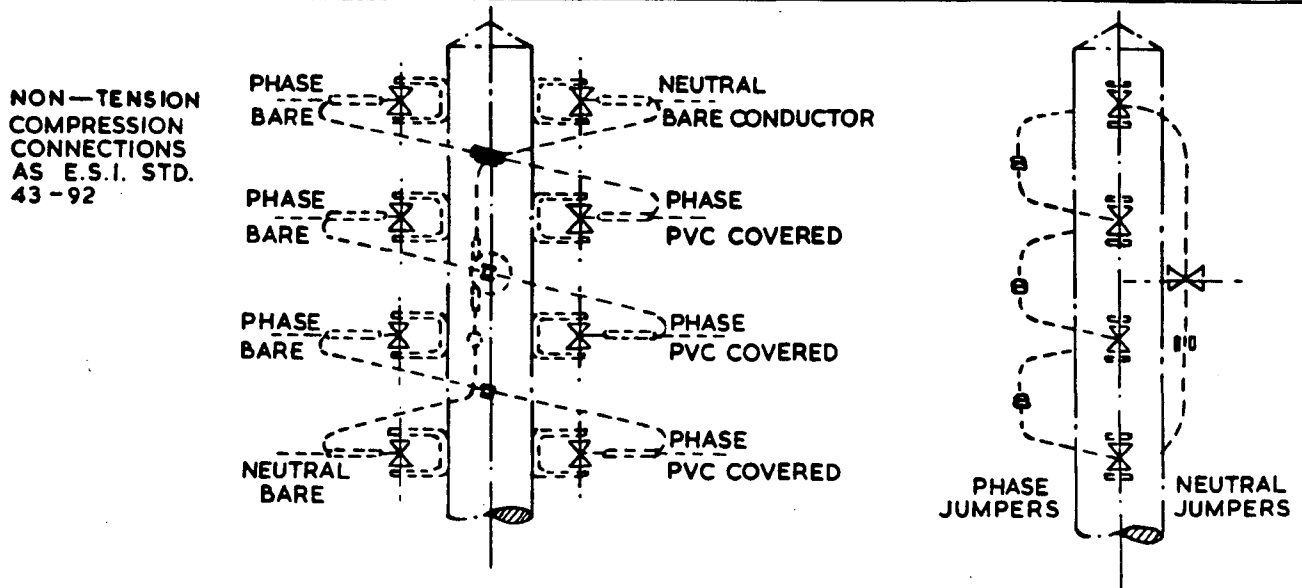
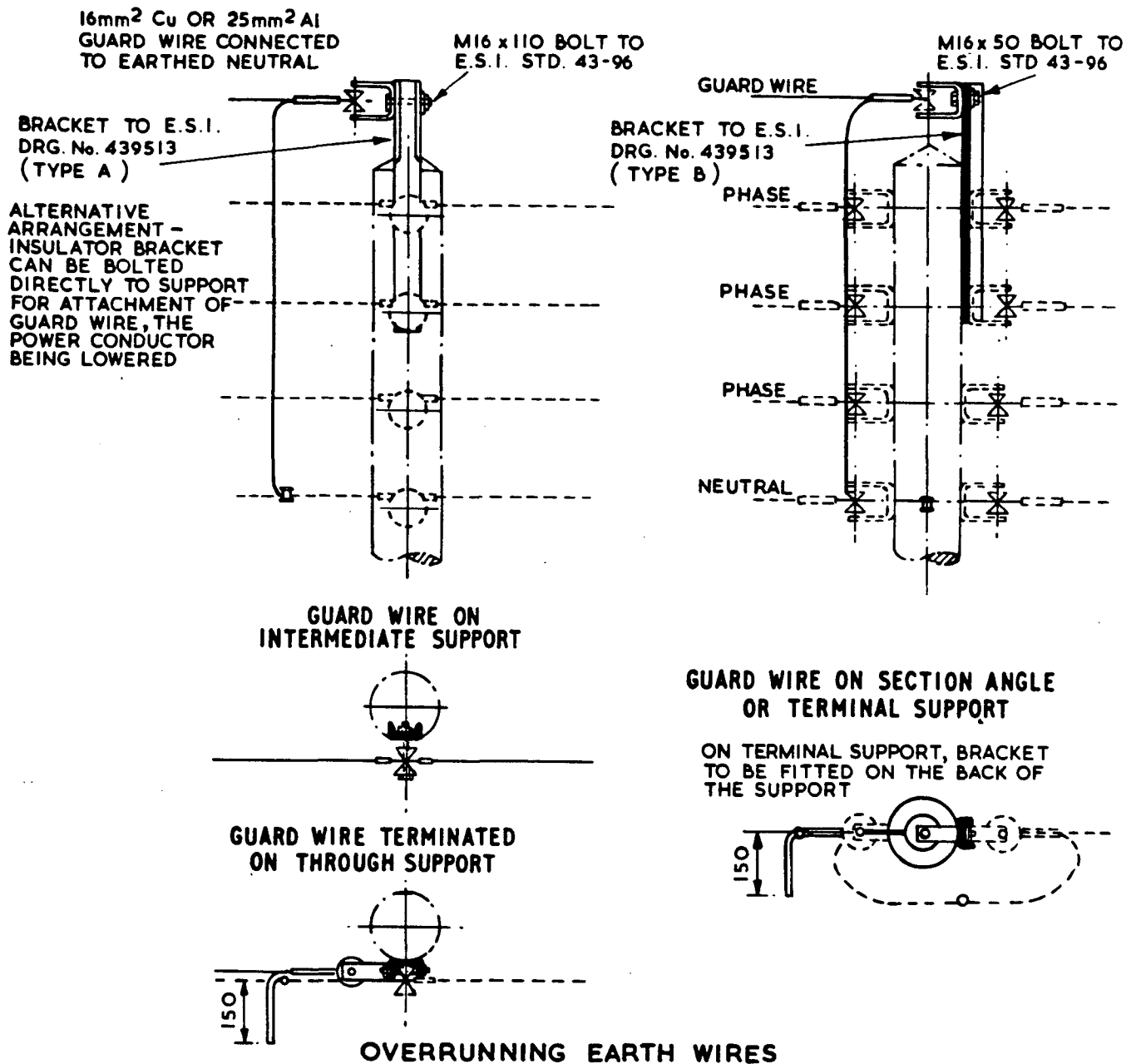


Fig. 21 Transposed neutral Post Office crossings undercrossing details
(ESI Drawing 433021)

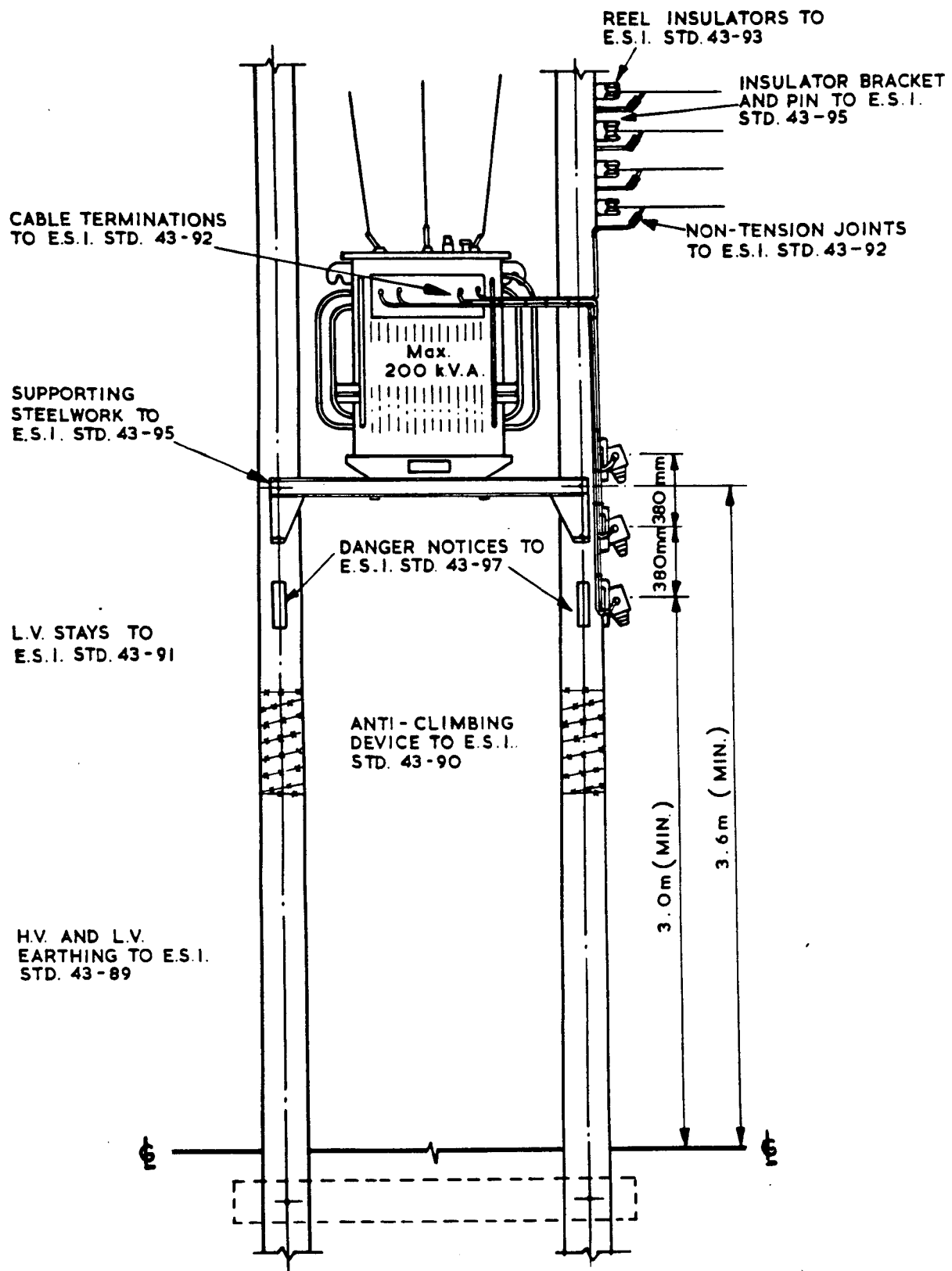


Fig. 22 L.V. transformer connections and earthing details
(ESI Drawing 433022)

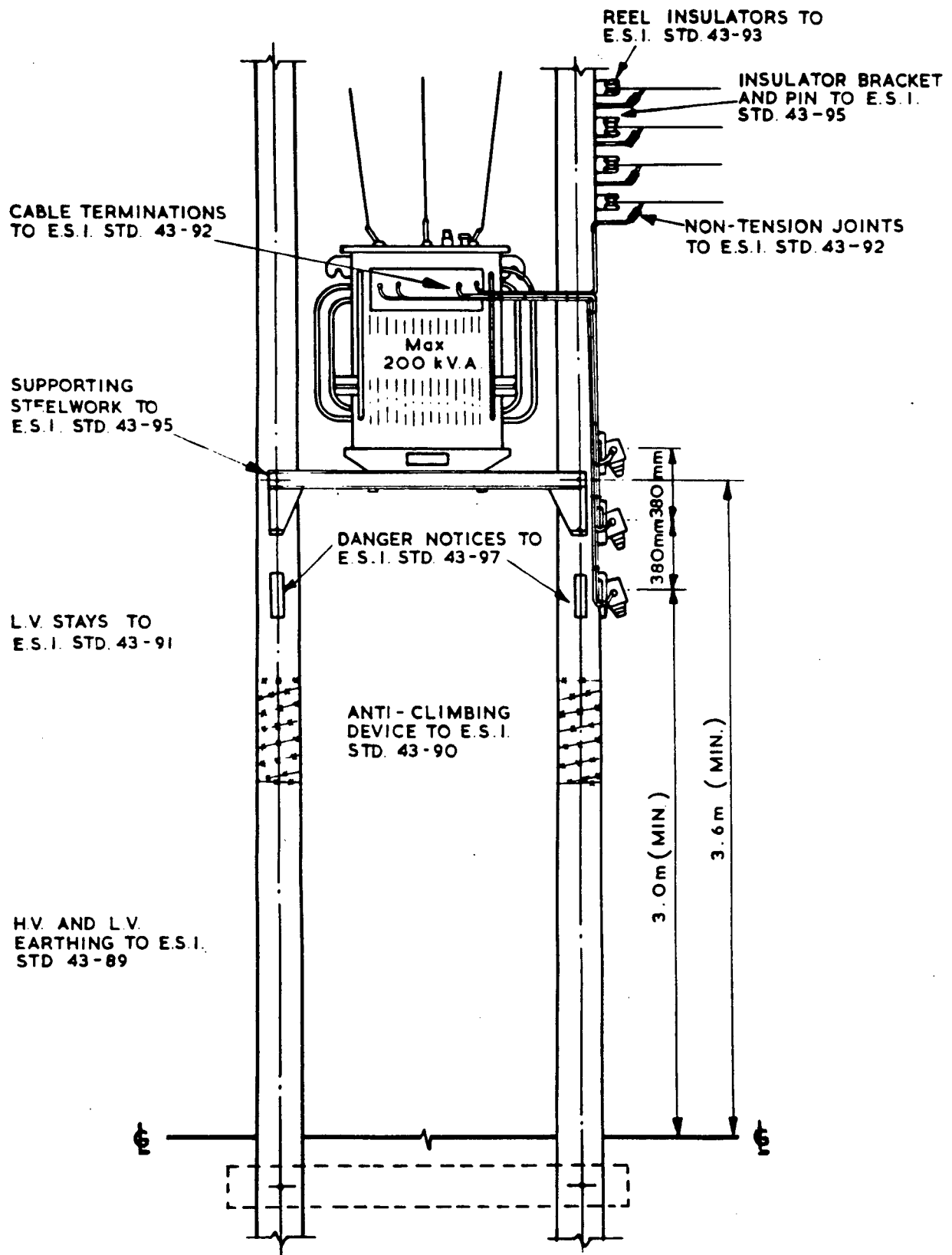


Fig. 22 L.V. transformer connections and earthing details
(ESI Drawing 433022)